

Model Driven Engineering, Artificial Intelligence, and DevOps for Software and Systems Engineering: A Systematic Mapping Study of Synergies and Challenges

LUCA BERARDINELLI*, Johannes Kepler Universität Linz, Austria

VITTORIANO MUTTILLO* and ROMINA ERAMO, University of Teramo, Italy

HUGO BRUNELIERE, IMT Atlantique, LS2N (CNRS), France

ABBAS RAHIMI, Johannes Kepler Universität Linz, Austria

ANTONIO CICCETTI, Mälardalens University, Sweden

JOAN GINER-MIGUELEZ, Barcelona Supercomputing Center, Spain

ABEL GÓMEZ, Universitat Oberta de Catalunya, Spain

PASQUALINA POTENA and MEHRDAD SAADATMAND, RISE Research Institutes of Sweden, Sweden

This paper presents a systematic mapping study classifying existing scientific contributions on synergies of Model Driven Engineering (MDE), Artificial Intelligence/Machine Learning (AI/ML), and DevOps, with the overall objective of supporting the continuous development of Cyber-Physical Systems (CPSs). We collected papers from bibliographic sources and selected primary studies to analyse. Then, we characterised and classified the current state of the art, focusing on 1) main aspects already tackled at the intersection of at least two of the three studied areas, and 2) findings emerging from the analysis as a framework for potential future research, notably regarding the integration of the three studied areas. The results reveal that few approaches combine MDE, AI/ML, and DevOps for software and systems engineering. In contrast, several approaches have combined two of them, specifically MDE and DevOps. Approaches combining AI/ML with MDE or DevOps are also becoming more frequent and will most likely continue to progress in the future. These synergies cover a range of engineering activities, from requirements and design to monitoring, maintenance, and evolution. Open research challenges include advancing AI/ML, MDE, and DevOps integration, supporting scalable, data-oriented solutions, proposing new continuous engineering methods, and adapting DevOps practices to diverse systems.

CCS Concepts: • **Computer systems organization** → **Embedded and cyber-physical systems**; • **Computing methodologies** → **Artificial intelligence**; **Machine learning**; • **Software and its engineering** → **Software creation and management**.

*Luca Berardinelli and Vittorioano Muttillio contributed equally to this research.

Authors' Contact Information: Luca Berardinelli, luca.berardinelli@jku.at, Johannes Kepler Universität Linz, Linz, Austria; Vittorioano Muttillio, vmuttillio@unite.it; Romina Eramo, reramo@unite.it, University of Teramo, Teramo, Italy; Hugo Bruneliere, hugo.bruneliere@imt-atlantique.fr, IMT Atlantique, LS2N (CNRS), Nantes, France; Abbas Rahimi, abbas.rahimi@jku.at, Johannes Kepler Universität Linz, Linz, Austria; Antonio Cicchetti, antonio.cicchetti@mdu.se, Mälardalens University, Västerås, Sweden; Joan Giner-Miguel, jginermi@uoc.edu, Barcelona Supercomputing Center, Barcelona, Spain; Abel Gómez, agomezlla@uoc.edu, Universitat Oberta de Catalunya, Barcelona, Spain; Pasqualina Potena, pasqualina.potena@ri.se; Mehrdad Saadatmand, mehrdad.saadatmand@ri.se, RISE Research Institutes of Sweden, Västerås, Sweden.

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1 Introduction

Nowadays, modern systems in the domains of Industry 4.0, healthcare, autonomously driving cars, and smart grids are examples of highly communicating systems in which software enables increasingly advanced functionality. The growing complexity of these systems (e.g., Cyber-Physical Systems (CPS) [60], Cyber-Physical System of Systems (CPSoS) [12]) poses several challenges throughout all system design, development, and analysis phases, but also during their deployment, actual usage, and future maintenance.

With the advent of DevOps principles [24, 37], system engineering would benefit from supporting a continuous development involving a smooth continuum of information from design to runtime and vice versa. Moreover, many leading companies have started envisioning the productivity boost of tomorrow to be brought by the application of full-blown Artificial Intelligence (AI) principles and techniques, including Machine Learning (ML) [53]. Through AI for IT operations (AIOps) [20], we can rethink the DevOps pipeline through continuous monitoring, alerting, and remediation in a secure and reliable way.

In this context, Model Driven Engineering (MDE) [13, 57] has gained popularity in software development projects as it raises the level of abstraction and thus improves the ability to handle the complexity of systems. Using models as purposeful abstractions of systems and their environment is a pillar enabling technology in current and future system engineering platforms [60].

However, to the best of our knowledge, although its relevance is recognized, few approaches intend to support full continuous software and system engineering processes integrating MDE, AI/ML¹, and DevOps altogether. Efficiently combining these three areas is a major objective of the AIDoArt large collaborative project [1, 15]. This three-year European effort, involving 31 academic and industrial partners from seven countries, aims to support the AI-augmented automation of modelling, coding, testing, monitoring, and continuous development in CPSs.

This paper reports on a Systematic Mapping Study (SMS) of the existing literature [50] we performed to explore the current state of both research methodologies and practical solutions that rely on different synergies between MDE, AI/ML, and DevOps. Thus, given the extensive body of work in the three areas, we focused on academic papers that discuss and consider a synergy of at least two of these areas. As a result, the main contributions of the paper are:

- An up-to-date cartography of the existing literature that classifies approaches in our context concerning (i) investigated areas and application domains; (ii) addressed problems and available tools; (iii) degree of proposed automation support;
- Based on this, a more detailed analysis of the potential innovation and future challenges to be faced at the intersection of the three studied areas, notably regarding a more complete integration of MDE, AI/ML, and DevOps altogether.

The rest of the paper is organized as follows. Section 2 provides information about the three pillar areas, i.e., MDE, AI/ML, and DevOps. Section 3 describes the protocol adopted to perform the SMS. Section 4 presents the SMS results and subsequent discussion. Section 5 evaluates the quality and limitations of the current SMS. Section 6 presents conclusions.

¹We use the term ‘AI/ML’ to encompass the full spectrum of AI approaches, including contributions based on ML techniques as well as those employing symbolic AI, heuristic methods, and other strategies not strictly classified as ML.

2 Background

Model Driven Engineering (MDE), Artificial Intelligence/Machine Learning (AI/ML), and DevOps are the three main research areas that together represent the *context* of this study. The following provides a brief background of these research areas.

2.1 Basic concepts and their relationships

Over the last two decades, MDE has become increasingly popular in software development projects as it raises the level of abstraction, thus improving the ability to handle complex systems (e.g., CPSs and CPSoSs). Using models as purposeful abstractions of systems and their environment is increasingly important for industrial applications [62]. The first-generation MDE tools primarily focused on generating code from high-level platform-independent abstract descriptions [56]. Since then, the target scope of MDE has broadened and now addresses, for example, model-based testing, verification, measurement, tool interoperability, software evolution, and many more hard key challenges in software engineering.

In such a context, [9] presents a model-based approach to support variable products in a system development environment. [SP49] presents an approach to maintain consistency between design models, the system under test, and test components. In addition, [54] uses a model-based methodology for model transformations to enable verification of system models using the AADL modelling language. [39] introduces a methodology to help developers maintain consistency in industrial model-based development projects by identifying inconsistencies during system development and maintenance. [63] examines issues and challenges that drive organizations to adopt model-based systems engineering as a solution to address the challenges of developing complex systems.

MDE benefits include (i) providing better abstraction principles and techniques, (ii) facilitating activities automation, and (iii) supporting technology integration among all the covered design and development activities. The heterogeneous data resulting from different engineering activities can be converted and then manipulated as MDE artefacts, contributing to making explicit the implicit modelling as performed in other project areas. By switching to model-driven approaches, businesses reap benefits through increased productivity and software quality [57].

Artificial Intelligence can be defined as a branch of computer science concerned with building intelligent machines [45] capable of finding solutions to complex problems in a more human-like fashion. AI research is highly specialized and divided into different areas such as machine learning, automated reasoning, knowledge-based systems, optimization, and natural language processing, which have successfully been applied in several industrial applications to achieve increased profitability, efficiency, safety, and security [2].

AI and ML technologies are increasingly used in day-to-day applications, affecting existing development practices by reducing the time-to-market and improving the quality of the software in general. The dissemination of such principles and techniques in a regulated industry can rapidly enable systems to decide and act in a more and more automated manner, sometimes even without direct human control [11].

Many AI and ML techniques have been used in several software engineering tasks. For example, in the requirements stage, requirements have been extracted from user stories and natural language requirements using natural language processing techniques [52]. During the development stage of the applications, machine learning techniques have facilitated code generation, documentation, and management through code completion, code search assistance, and automating code commenting and documentation [64]. Besides, numerous machine learning-powered software testing and debugging tools are intended to find and address bugs already occurring during the development of software systems and to aid in the test generation and during the execution of these tests [68]. Regarding the post-deployment phases of software systems, several research projects have also used machine-learning techniques to aid in software maintenance. Their proposed techniques are to identify cloned code, suggest changes to the code in every version release, and categorize user reviews automatically. [27]

DevOps is a software development approach that bridges the traditional divide between development (Dev) and operations (Ops) teams in an organization [37, 42, SP96]. The primary goal of DevOps is to shorten development and deployment cycles between Dev and Ops based on a more collaborative relationship and agile collaboration between both factions. It enables continuous feedback and quick response to changes and uses automated delivery pipelines, reducing cycle time. To this end, DevOps introduces automation into all stages of the software development lifecycle—from integration, testing, and release to deployment and infrastructure management. As a result, software development and evolution become more efficient and reliable.

Artificial Intelligence for IT Operations (AIOps) is the most prominent synergy between DevOps. The term AIOps was coined by Gartner [8, 53] to characterize solutions where DevOps challenges are addressed with the help of AI/ML techniques. A fundamental challenge resides in integrating AIOps [20] in enhancing the DevOps pipeline, for example, through continuous alerts and insights from data used for continuous deployment and operations management. AIOps uses data science and computational techniques to automate common and routine operational tasks, digest metrics, and use inference models to extract actionable insights from data.

On the one hand, AI/ML is a relevant complementary means, along with MDE, providing more efficient and flexible support to the continuous development of complex systems, such as CPSs [70]. On the other hand, the DevOps toolchain can be enhanced by employing AI techniques, in particular ML, in multiple aspects of the system engineering process (design and modelling quality, source code performance, testing coverage, and predictive monitoring, to name a few) [43, SP63].

Data represents a key aspect of the synergy of MDE, AI/ML, and DevOps is concerned [69]. The information recovered from a given system execution, either historical data or real-time traces (such as car sensor data or radar-based traffic monitoring data), has a potentially very high value in assessing the validity and required properties of a system design. However, dedicated methods and tools are needed to extract knowledge from large volumes of such runtime data and analyze and (re)use it in a productive manner. Data science analysis draws insights from structured and unstructured data (i.e., data, raw data, metadata, ...) [71]. To this end, integrating data-related technological innovations can ensure that systems are designed responsibly in a way that contributes to our trust in their behaviour.

Thanks to the application of advanced MDE principles and techniques, the heterogeneous data resulting from different engineering activities can be converted and then manipulated as MDE artefacts (i.e., models, metamodels), contributing to making explicit the implicit modelling as performed in other project areas. All data from these heterogeneous sources can be processed, collected, traced, and analyzed more homogeneously (e.g., relying on common shared repositories and model views [14]). This way, relevant data can be made available to all the AI-based and/or DevOps-based services that require it.

As the purpose of this work, the study of the synergies among MDE, AI/ML, and DevOps can help researchers and practitioners to develop scalable solutions for efficient handling and management of numerous, heterogeneous, and large models potentially covering several functional and non-functional perspectives of the system under development (in a DevOps process) suitable for large distributed cross-functional working teams and allowing to integrate feedback (notably from runtime data) to the system level models [SP101].

2.2 Related work

To the best of our knowledge, no existing systematic mapping studies consider the synergies between MDE, DevOps, and AI/ML to the extent of this article. Nonetheless, several secondary studies focus on subsets of our investigation or address specific research fields that emerged as a consequence of such synergies.

In general, DevOps adoption poses significant challenges both from an organizational and a technical perspective: Laukkanen et al. [40], Shahin et al. [58], and Leite et al. [42] describe the results of systematic studies on the adoption DevOps and Continuous Development/Integration/Deployment practices. While these works shed light

on recurring issues and possible solutions when pursuing continuous processes, they do not address the specific problems and potential of including MDE and AI/ML in the context.

MDE is often conceived as a resource and a hindrance in adopting DevOps and, generally, continuous processes [6]. Correspondingly, surveys related to DevOps and continuous processes often include MDE methodologies as a fundamental ingredient, notably for automated deployment [67] or continuous testing [51]. On the other hand, MDE solutions might suffer (frequent) changes, e.g., to keep traceability links updated, to detect and manage distributed modifications, and generally, to handle consistency needs [6]. Related to consistency management, digital twins have emerged as a suitable technology to support the feedback loop from operations to development: Djebali et al. [22] present a survey on continuous digital twins design with a particular focus on the smart-grid domain of application. In our SMS, we investigate the cases where MDE supports DevOps and the literature in which DevOps processes are adopted to realize continuous MDE.

AI/ML solutions are becoming ubiquitous as components of a target system and as technology to improve the development process. Martínez-Fernández et al. distinguish between SE4AI (Software Engineering for AI) and AI4SE (AI for SE, also known as intelligent SE) and present an SMS on SE4AI [44]. Moreover, Perkusich et al. [48] discuss the results of a survey on AI4SE. Both studies agree on the infancy of these research fields, even though it is expected to be a burst of new research results due to the recent advent of Large Language Models. Interestingly, in a recent survey by Diaz-de-Arcaya et al. [21], the distinction between SE4AI and AI4SE becomes blurrier when considering DevOps processes. These cases are referred to as AIOps or MLOps, and the AI/ML mechanisms are both part of product components and contribute to the feedback loop for improving the product itself. This line of reasoning is consistent with the goals of our SMS, which can be considered a superset of the investigation in [21] since we are investigating the synergies and cross-fertilization opportunities between AI/ML and SE processes, resources, and products. Naveed et al. [46] delve into the intersection of MDE and machine learning components, which they term MDE4ML. Their study aims to analyze existing research on MDE4ML (a subset of MDE4AI). The authors highlight the increasing adoption of ML as a component in various software applications, a trend driven by the desire to harness large volumes of data for predictive capabilities and informed decision-making.

3 Method

To understand state-of-the-art research practices on MDE, AI/ML, and DevOps to support the system engineering process we conducted a systematic mapping study (SMS) of the existing literature. An SMS is a method for aggregating systems and software engineering research. It provides a structure of the research results that have been published by categorizing them and giving a visual summary of the results. This SMS is based on the guidelines proposed by Petersen et al. [49, 50], and the *snowballing* process defined by Wohlin [65].

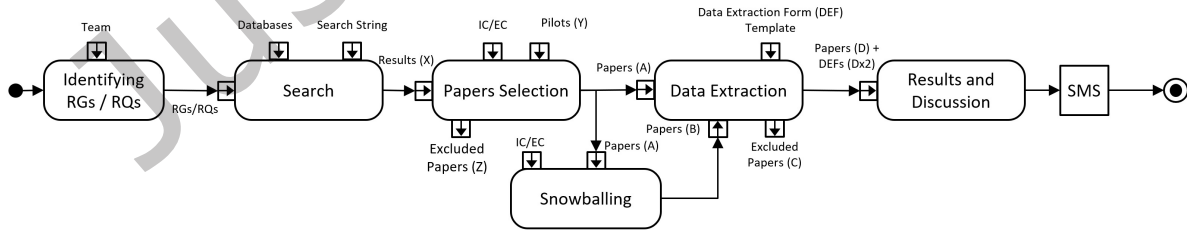


Fig. 1. The protocol workflow.

The protocol workflow is depicted using an activity-like diagram in Figure 1. It consists of the following steps:

- *Identifying the Research Goals (RGs) and the Research Questions (RQs)*: The objective of this work and the research questions are defined by the SMS team.
- *Initial Search*: the literature search is conducted on selected databases, collecting scientific publications by executing queries based on a search string suitably designed according to the given RGs and RQs.
- *Pilot Papers Collection*: a set of *Pilot Papers* has been collected. They represent a set of papers that are considered of interest for/related to the research areas of this study and are matched against the collected results to assess the effectiveness of the designed search string(s)²
- *Papers Selection*: Inclusion and Exclusion Criteria (IC/EC) are defined and applied to the retrieved papers by considering their titles and abstracts.
- *Snowballing*: This process involves two steps, namely *backward* and *forward* snowballing. *Backward* snowballing is the process of reviewing the references cited in the selected studies, essentially looking at works published before the study in question. On the other hand, *forward* snowballing involves examining the documents that have cited the selected study, essentially focusing on works published after the study.
- *Data extraction*: Once a collection of papers is obtained by appropriately applying the defined IC/EC, comprehensive data extraction is done by examining the full text of each selected paper. A Data Extraction Form (DEF) is utilized by reviewers to systematically gather detailed information from these papers, aiding in the resolution of the Research Questions (RQs). The application of IC/EC typically persists throughout a full reading of the papers, and if necessary, some papers may be further excluded.
- *Results Analysis and Discussion*: The gathered results are scrutinized, and a collaborative discussion is held among the team members. This discussion is documented in the SMS to address the proposed RQs. The DEFs, which each reviewer has filled out, are consolidated to compile comments and to categorize the selected studies appropriately.

The protocol execution is documented in a spreadsheet collaboratively edited by the involved partners [3]. The following Sections separately describe the protocol steps and activities in detail.

3.1 Team

We rely on a team of experienced members with heterogeneous MDE, AI/ML, and DevOps competencies to execute the protocol. It included members who were experts in at least one of the dimensions considered in the SMS coming from academia/research organizations, SMEs, and large industries. The team was formed within the European project AIDoArt (AI-augmented automation for efficient DevOps, a model-based framework for continuous development At RunTime in cyber-physical systems) [1, 15, 26]. AIDoArt focuses on supporting the continuous development of CPSs via AI-augmented automation, incorporating methods and tools for continuous software, system engineering, and validation.

3.2 Research Goals and Research Questions (RG/RQ)

Table 1. Research goal

<i>Purpose</i>	To identify and classify existing solutions
<i>Issue</i>	leveraging the synergies between MDE, DevOps, and AI/ML principles and practices
<i>Object</i>	supporting the system and software engineering of cyber-physical systems
<i>Viewpoint</i>	from the point of view of researchers and practitioners.

²The higher the number of matches, the better is/are the chosen search string(s).

The overall Research Goal (RG) of this SMS, defined using the Goal-Question-Metric perspective [7], is given in Table 1. This study examines scientific publications that discuss three key areas: MDE, AI/ML, and DevOps. It aims to report on solutions that effectively integrate these three areas. Starting from such RG, we formulated the following Research Questions (RQ):

- *RQ1*: Is there a systems and software engineering methodology that explicitly incorporates and integrates the principles and practices of MDE, AI/ML, and DevOps research areas? If such a methodology exists, how does it combine these research areas?
- *RQ2*: Are the principles and practices of MDE, AI/ML, and DevOps research areas integrated throughout the entire process, or are they applied to specific engineering activities?
- *RQ3*: Which research fields and application domains are the target of these approaches?
- *RQ4*: What are the future research directions?

3.3 Search

In this section, we describe in detail the search process performed in the protocol in Fig 1. In particular, it starts with the design of the search string and the identification of reference databases.

3.3.1 Search Strings Table 2 summarizes the resulting search string. It is formulated as the conjunctive logic proposition of three sets of search terms, which, in turn, are combined in a disjunctive logic proposition. Since the integration of MDE, AI/ML, and DevOps principles and practices applied to CPS engineering is the scope of this work, a search string is required that returns only those combinatorial interaction strategies that support string constraints. In doing so, these search strings were evaluated based on the extent to which the returned results are related to the application domain. To ensure that the search string did not omit relevant papers, *pilot papers* (discussed in the next section) were selected to ensure that the search string could find these core/pilot studies. The experiment translated the database-independent proposition in Table 2 in the database-specific query string(s). The database-specific strings are available in [3] for the sake of future SMS updates³.

Table 2. Search strings

Keyword	Searching String
MDE	{Low-code} OR {MBE} OR {MBSE} OR {MBT} OR {MDA} OR {MDD} OR {MDE} OR {MDSE} OR {Model-Based Engineering} OR {Model-Based Requirement Engineering} OR {Model-Based Requirement Specification} OR {Model-Based Software Development} OR {Model-Based Software Engineering} OR {Model-Based System Engineering} OR {Model-Based Testing} OR {Model-Driven Architecture} OR {Model-Driven Development} OR {Model-Driven Engineering} OR {Model-Driven Software Architecture} OR {Model-Driven Software Development} OR {Model-Driven Software Engineering} OR {Model-Driven System Architecture} OR {Model-Driven System Development} OR {Model-Driven System Engineering} OR {Model-Driven Testing} OR {Runtime Model} OR {Digital Twin} OR {Digital Twinning}
AI/ML	{AI} OR {Artificial Intelligence} OR {Big Data} OR {Big Data Analytics} OR {classification} OR {clustering} OR {data analytics} OR {data mining} OR {data science} OR {decision trees} OR {Deep Learning} OR {linear model} OR {Machine Learning} OR {ML} OR {natural language processing} OR {Neural Network} OR {NLP} OR {regression} OR {reinforcement learning} OR {search based} OR {supervised learning} OR {support vector machines} OR {unsupervised learning}
DevOps	{DataOps} OR {DevOps} OR {AIOps} OR {DevSecOps} OR {Continuous delivery} OR {continuous integration} OR {continuous software engineering} OR {continuous development} OR {continuous engineering} OR {continuous system engineering} OR {continuous deployment} OR {continuous quality} OR {continuous testing}
CPS	{Cyber-Physical System} OR {Cyber-Physical Systems} OR {CPS}

³SMS Additional Material: <https://hdl.handle.net/20.500.12004/1/U/358195>

3.3.2 Databases Considering the interdisciplinary nature of the SMS, we performed an automatic search on leading research and rich databases in order to provide adequate literature coverage. The databases were selected based on the guidelines and suggestions provided by [23, 50]. Based on these guidelines, the databases consulted are as follows:

- *Scopus* (<http://www.scopus.com/>): it is a scientific and interdisciplinary bibliography database created by Elsevier, which contains abstracts and citations for academic journal articles. We decided to use Scopus since it is a meta-library and indexes publications by well-known and reputable publishers, such as ACM, IEEE, Springer, and Elsevier.
- *ACM Digital Library* (<http://dl.acm.org/>): it is one the most relevant academic databases primarily addressed to researchers in the field of computer science. Even though Scopus indexes the ACM Digital library, we decided to include it to ensure that no publications were excluded from Scopus' indexing of ACM publications.
- *IEEE Xplore Digital Library* (<http://ieeexplore.ieee.org/>): it is the most well-known resource for discovering and accessing the scientific and technical content published by the IEEE (Institute of Electrical and Electronics Engineers) and its publishing partners. Although Scopus covers IEEE Xplore, it was included to verify the quality of the search strings.
- *DBLP Computer Science Bibliography* (<http://dblp.uni-trier.de/>): it has become a relevant resource in the field of computer science. We selected the DBLP database since it represents one of the most relevant sources in the computer science landscape.
- *SpringerLink* (<https://link.springer.com/>): it is a scientific database operated by Springer International Publishing, which represents a powerful central access point for Science, Technology, and Medicine researchers. The SpringerLink database was selected based on expert opinion and to ensure good coverage in the area of computer science and engineering.

During searching, indexing, and sorting a large number of references, different duplicate references appeared due to the slight differences in the database indexing. To manage the references and to remove duplicates, the obtained results have been cleaned with Tidy Bib Web Tool ⁴ and have been exported in BibLaTeX (.bib), imported in a shared Zotero ⁵ web library ⁶ [4], to remove duplicates, exported (e.g., as CSV documents) and imported in Google Sheet for convenient collaborative sorting/filtering and documentation purposes ⁷.

3.4 Papers Selection

This step starts with *i*) a set of unique papers collected from the aforementioned databases, *ii*) a set of potential pilot papers based on individual backgrounds of team members, and *iii*) a set of IC/EC to be applied to these two sets of documents. The paper selection is detailed in the following subsections.

3.4.1 Inclusion and Exclusion Criteria (IC/EC) We defined inclusion and exclusion criteria that, as the name suggests, determine whether a paper is part of the SMS or must be excluded.

IC/EC have been evaluated for each paper collected by queries executed on the selected databases by reading its title and abstract (T/A) or, in addition, the full text (All), as reported in Table 3.

3.4.2 Pilot Papers Selection Based on the stated RG/RQ, and once agreed IC/EC, we collected potential pilot studies, i.e., interesting documents that, based on the individual expertise of authors, could be inspirational and/or related to the running SMS.

⁴Online BibTeX Tidy: <https://flamingtempura.github.io>

⁵Zotero (<https://www.zotero.org/>) is a free and open-source reference management software to manage bibliographic data and related research materials (such as PDF files).

⁶The web library is kept private and accessible by invitation and made available after the publication of the SMS in venues or journals

⁷The DOI is used as key to import/export entries to/from Google Sheet and Zotero.

Table 3. Inclusion and exclusion criteria

Criteria	Assessment Criteria	Step
Inclusion	Challenges paper	T/A
	Book chapters	T/A
	Tool papers mentioning at least 2 of 3 research areas (MDE, AI/ML, DevOps)	All
	Papers from high ranked journals (Q1/Q2 Scimago) and conferences (GRIN-GII-SCIE)	T/A
	Papers published after 2009	T/A
Exclusion	Non-scientific publications / grey literature	T/A
	Books in their entirety	T/A
	Tool papers mentioning only 1 of 3 research areas (MDE, AI/ML, DevOps)	All
	Project reports	T/A
	Summaries of workshops/conferences	T/A
	Bachelor Master and PhD theses	T/A
	Study with data not reliably extracted, duplicate, or overlapping data	T/A
	Abstract-only papers as WIP papers, conference, reviews, and editorial	T/A
	Papers without available full text available	T/A
	Non-peer-reviewed papers, case reports, case series, technical reports, and web pages	T/A
	Papers from low ranked journals (higher than Q1/Q2 Scimago) and conferences (GRIN-GII-SCIE)	T/A
	Letters	T/A
	Secondary studies and interviews	T/A

Table 4. Pilots Papers

Reference	Title (Pilots retrieved by the search query in <i>italics</i>)	Year
[SP100]	<i>Model-Driven Continuous Deployment for Quality DevOps</i>	2016
[SP147]	<i>The Next Evolution of MDE: A Seamless Integration of Machine Learning into Domain Modeling</i>	2017
[SP47]	<i>Cognifying Model-Driven Software Engineering</i>	2018
[SP131]	<i>Stepwise Adoption of Continuous Delivery in Model-Driven Engineering</i>	2019
[SP171]	<i>When DevOps Meets Meta-Learning: A Portfolio to Rule them all</i>	2019
[SP63]	<i>DevOpsML: Towards Modeling DevOps Processes and Platforms</i>	2020
[SP101]	<i>Model-Driven ML-Ops for Intelligent Enterprise Applications: Vision, Approaches and Challenges</i>	2020
[SP158]	<i>Towards Integrating Data-Driven Requirements Engineering into the Software Development Process: A Vision Paper</i>	2020
[SP160]	<i>Towards Modeling Framework for DevOps: Requirements Derived from Industry Use Case</i>	2020
[SP3]	<i>A Hitchhiker's Guide to Model-Driven Engineering for Data-Centric Systems</i>	2021

Each paper from such a list underwent (at least one) individual assessment by team members who applied the IC/EC. The result was a curated list of pilots shown in Table 4.

3.4.3 Papers Selection Procedure The selection procedure for collected papers consists of the following steps:

- *Assess the quality of the search strings.* The quality of the search string is quantified by counting how many pilots are retrieved by the search query execution on search databases,
- *Search.* All the search strings are executed on the selected databases,
- *Search results.* Search results are collected (e.g., as BibTeX entries) in a reference database (e.g., Zotero [4]),
- *Applying IC/EC to title and abstract.* For all collected studies, IC/EC are applied. Each collected study is reviewed by at least two persons. The review at this stage consists of reading the title and abstract only. Each reviewer collects her/his IC/EC evaluations in a reviewer table and, eventually, all the reviews are integrated,
- *Conflict management.* An agreement/disagreement table (ADT) is required for conflicting reviews. An additional reviewer is then assigned to each conflicting study and a final decision is taken.

Table 5. Data extraction form

ID	Question	Outcome	RQ(s)
Q1	The paper can be classified as MDE for DevOps?	{Yes, No}	RQ1
Q2	The paper can be classified as MDE for AI/ML?	{Yes, No}	RQ1
Q3	The paper can be classified as DevOps for MDE?	{Yes, No}	RQ1
Q4	The paper can be classified as DevOps for AI/ML?	{Yes, No}	RQ1
Q5	The paper can be classified as AI/ML for MDE?	{Yes, No}	RQ1
Q6	The paper can be classified as AI/ML for DevOps?	{Yes, No}	RQ1
Q7	The paper can be classified as MDE + AI/ML + DevOps?	{Yes, No}	RQ1
Q8	Related Use Case Scenario(s)	{Requirements, Modelling, Monitoring, Coding, Testing, Others [30]}	RQ2
Q9	Keywords	List of keywords	RQ2
Q10	PaperType + Journal/Conference/Workshop Name	{Journal, Conference, Workshop, Book Chapter} + Publication details	RQ3
Q11	Application Domain(s)?	{Automotive, Aerospace, Railway, Maritime, Construction, Digital Life, Manufacturing, Application domain independent}: at least one, up to five or domain independent	RQ3
Q12	Case Study / Example availability	Number of available case studies, additional description/details	RQ3
Q13	Tools Availability	{Yes, No}	RQ3
Q14	Should the paper still be included in the SMS?	Field automatically calculated	
Q15	Future research directions (as stated by authors, if any)	Text manually extrapolated from the paper	RQ4

3.5 Data Extraction

To answer our RQs, we extracted different types of data. First, we used a predefined data extraction form template to review the full-text papers. Furthermore, to support our discussion and extract topics of interest, we systematically analysed keywords, extracted semantic/ontological information, and utilised NLP methods to validate the results obtained.

3.5.1 Paper Review In this phase, each paper from the set of selected studies underwent a detailed review requiring a full read. Two different reviewers reviewed each paper.

Each reviewer has been provided with a data extraction form template with guidelines suggesting whether or not a selected study contributes to answering the stated RQs. Table 5 shows the structure of the data extraction template.

To answer the RQs, we extracted general data related to the contributions of the papers, case studies and tools availability, publication data, application domain, and keywords (i.e., Q1-Q15). The data extraction template also provides support⁸ to suggest whether the paper should (not) be considered part of SMS (i.e., Q14). In particular:

- To answer the RQ1 (what approaches combining MDE, DevOps, and AI/ML have been proposed), we considered the approaches that cover and combine at least two of the three areas. In particular, Q1-Q7 are considered.
- To answer the RQ2 (what software engineering activities and perspectives have been considered), we gave thought to the type of use case scenarios (Q8), and refined the analysis by considering the collected keywords (Q9).
- To answer the RQ3 (what fields and application domains have been considered), we examined the publication venue (Q10), application domains (Q11), and case studies (Q12) along with available tools/frameworks (Q13).
- To answer the RQ4 (future research directions), we considered the answers to the Q15.

3.5.2 MDE, DevOps, and AI/ML Body of Knowledge To answer RQ1, we have considered the Body of Knowledge (BoK) for MDE, AI/ML, and DevOps as the basis for our classification⁹. In this section, we present the BoK and explain how it supports our classification and mapping.

DevOps BoK: Jabbari et al. [38] categorize DevOps practices according to the **Software Engineering Body of Knowledge (SEBoK)**. These knowledge areas provide a structured framework for aligning DevOps practices with core software engineering concepts. For instance, practices such as continuous planning, feedback loops, and continuous monitoring fall under **Software Engineering Management**, reflecting the emphasis on managing and improving workflows between development and operations. **Software Construction** includes practices like continuous integration and prototyping applications, which are essential for iterative development. **Software Configuration Management** focuses on practices such as continuous deployment, automated deployment, and change management, ensuring smooth transitions from development to production environments. **Software Testing** emphasizes continuous testing and automated testing to streamline quality assurance and reduce feedback cycles, while **Software Process** defines and standardises processes to support production workflows. In **Software Quality**, practices focus on using data and metrics to guide quality assurance. **Software Engineering Tools and Methods** cover practices like infrastructure as code, performance metrics, and modeling and simulation, which optimize and automate operations. **Software Requirements** address the collaborative definition and management of requirements, supporting iterative development. Finally, **Software Design** focuses on architectural design to ensure robust and scalable systems.

MDE BoK: Furthermore, Burgueno et al. [16] categorize topics, according to the **Model-Driven Software Engineering Body of Knowledge (MDEBoK)**, into key macro areas that represent foundational aspects of MDE. These include **Model Foundations**, which cover the core concepts of modeling, such as syntax, semantics, and modeling intents like meta-modeling and transformation definitions. **Model Quality** focuses on completeness, consistency, correctness, and comprehensibility to ensure models are fit for their purpose. **Analysis** includes structural, behavioral, and model transformation analysis, addressing techniques like invariant checking, metrics, and performance analysis. **Modeling Languages** explores the definition and usage of languages, including general-purpose and domain-specific types and multiview modeling approaches. **Model Representation** deals

⁸Via Google Sheet functions aggregating data entered by the reviewer.

⁹MDE, DevOps, and AI/ML BoK: <https://hdl.handle.net/20.500.12004/1/U/591875>

with concrete syntax, notations, layouts, and visualization techniques, encompassing both textual and graphical forms. **Model Maintenance and Evolution** addresses operations such as versioning, migration, and refactoring to manage model changes over time. **Model Execution** focuses on simulation, execution strategies, and debugging/testing of executable models. **Model Transformations** examine transformation languages, various types of transformations (e.g., text-to-model, model-to-model), and their applications like code generation, synchronization, and optimization.

AI BoK: Finally, Samoili et al. [18] categorize the core and transversal areas of artificial intelligence according to the AI taxonomy, outlining the **AI Body of Knowledge (AIBoK)** into structured domains and subdomains. The core domains include foundational AI capabilities such as 1) **Reasoning**, which involves knowledge representation and automated reasoning; 2) **Planning**, encompassing scheduling, searching, and optimization; 3) **Learning**, which covers ML techniques like supervised, unsupervised, and reinforcement learning, as well as neural networks and generative models; 4) **Communication**, focusing on natural language processing tasks like translation and sentiment analysis; 5) **Perception**, which deals with computer vision and audio processing for image and speech recognition. The transversal domains expand the AIBoK to include interdisciplinary applications and considerations, such as 6) **Integration and Interaction** in multi-agent systems, robotics, and connected vehicles; 7) **AI Services** provided through cloud platforms, software tools, and frameworks; 8) **Ethics and Philosophy**, which address transparency, fairness, accountability, and broader societal impacts of AI.

3.5.3 Keyword Analysis As part of the data extraction process, we collected the keywords of each reviewed paper. We used the analysis of these keywords to identify the key fields of interest in the selected papers (Q1-Q7). Before analyzing them, we first performed a syntactical cleaning followed by a normalization process:

- (1) We capitalized each word, except for articles and prepositions such as in “Internet of Things”;
- (2) We removed hyphens and special characters, except when widely used in the literature such as in “Cyber-Physical System” and “Model@Runtime”;
- (3) We replaced abbreviations with their literal counterparts, e.g. “MDE” is systematically replaced by “Model-Driven Engineering”.

3.5.4 Ontology Several enhancements have been made to the analysis process of the collected keywords. A major enhancement concerns the semantic relations between the keywords. First, equivalent terms have been identified semantically and not only syntactically. Furthermore, the collected keywords have been organized in a tree of correlated terms, like an ontology. To do this, we sketched a list of questions that ontology should be able to answer using the competency questions [35] approach (e.g., Is *Metamodeling* a subclass of *Model-Driven Software Engineering* or a child of *Model-Driven*). Next, we created a list of all terms that we would like to explain, and we developed a class hierarchy with a combination of top-down (general concepts into specializations) and bottom-up (specific classes into more general abstract concepts) approaches [61]. Once we defined Classes and SubClasses, we described their internal structure, namely the object properties of the concepts in these Classes, so that an object property becomes a Slot of this Class (i.e., a leaf in the ontology tree). The produced Ontology provides a basic structure for representing the relationships between the concepts found in the keyword list, focusing on the relationships among the provided keywords and capturing the relationships between concepts. The frequency of keywords metric, called *Weighted Keyword (WK)*, has been used to weight the Classes (i.e., the value *WK* represents a semantical weight assigned to each Class/Slot obtained through the Class/Slot word frequency in the keywords and the sum of the frequency of all SubClasses and Slots belonging to the considered Classes). Moreover, we also extracted relationships among the ontology research area classes (i.e., MDE, DevOps, Artificial Intelligence and Machine Learning), extracting the keywords appearing together with each dimension (Q1-Q7). For instance, we selected the keyword “Model-Driven Engineering” and retrieved all selected papers containing it as a keyword. Subsequently, we count the occurrences of all keywords associated with the considered

class (e.g., Model-Driven Engineering with other keywords related to DevOps or AI/ML research areas), linking them to the Ontology's Classes and Slots. These relationships and their weights (i.e., the cross-links among the Ontology Classes) aid in elucidating the connections between various SubClasses. For instance, MDE, DevOps, and AI/ML are distinct in the ontology (i.e., they belong to different tree paths). Still, in the selected papers, there exists an overlap of terms from these three areas, enabling us to quantify, at least in terms of keywords, the types of relationships between MDE, DevOps and AI/ML and their SubClasses (Q1-Q7). Furthermore, this work presents a coarse-grained analysis, employing generic keywords relevant to the considered research areas. Future work will refine this analysis by considering relationships between keyword frequencies and the Classes/Slots of Ontology, incorporating cross-relationships and classes based on domain knowledge (Q12).

3.5.5 Topic Analysis Topic modelling is an unsupervised machine learning technique that automatically discovers thematic structures within a large collection of text documents (i.e., the so-called “corpus”) [47]. Each document addresses one or more topics. Unlike supervised learning, topic modeling does not require pre-annotated data or training sets. The output consists of a set of “ k ” topics, each described by its most frequent terms and the relevance score of each document to that topic. A single document can belong to multiple topics.

In this work, we used a publicly available online tool based on Scikit-learn libraries [34] to simplify document-term matrix creation. According to the tool input's data type, the selected papers have been stored in a single file and formatted so that one paper appears on each line. The first step then involved tokenization (i.e., splitting text into individual terms or tokens). In English, this is done by splitting on whitespace. After preprocessing the text, where non-informative stopwords were removed, each paper was represented as a vector in an m -dimensional space, where m is the number of unique terms across all selected papers. This set of terms is called the corpus vocabulary. The number of times a term appears in a document determines its value in the vector. All documents are then transformed and stacked into a full document-term matrix. We further enhanced the document-term matrix by incorporating TF-IDF weighting, which assigns higher weights to more informative terms. TF-IDF consists of two functions: 1) *Term Frequency (TF)* that measures the frequency of a term within a single document, and 2) *Inverse Document Frequency (IDF)* that captures the term's overall importance across the corpus. The terms appearing in many documents receive lower weights. This weighting method emphasizes terms that are frequent within a document but rare across the corpus, leading to a more informative representation of documents, following the equation below:

$$\omega(t, D) = TF(t, d) \cdot IDF(t) = TF(t, D) \cdot \left(\log \left(\frac{n}{DF(t)} \right) + 1 \right) \quad (1)$$

where t is the considered term, D is the set of all documents (i.e., the corpus), d is the considered document, and n is the total number of documents. We employed Non-negative Matrix Factorization (NMF) with SVD-based initialization for topic modelling on the TF-IDF weighted matrix [10][41]. A crucial step involves selecting the number of topics “ k ”. We explored a range of k values [$k_{min} = 4, \dots, k_{max} = 40$] and evaluated topic coherence using the mean TC-W2V measure [47]. Moreover, we selected the k value that yielded the highest mean coherence. Finally, we cluster the topics according to ontology nodes and super-classes (Q1-Q7).

3.6 Classification framework

The resulting classification framework is shown in Figure 2 and it is composed of three main facets, each of them addressing its corresponding research question¹⁰.

- **Synergies between MDE, AI/ML and DevOps (RQ1):** After identifying the synergies between the three research areas *MDE*, *AI/ML* and *DevOps* through the paper review (described in Section 3.5.1), we examine

¹⁰It is worth noting that the framework adoption was made possible by our targeted search strings, which ensured a comprehensive and systematic retrieval of relevant literature.

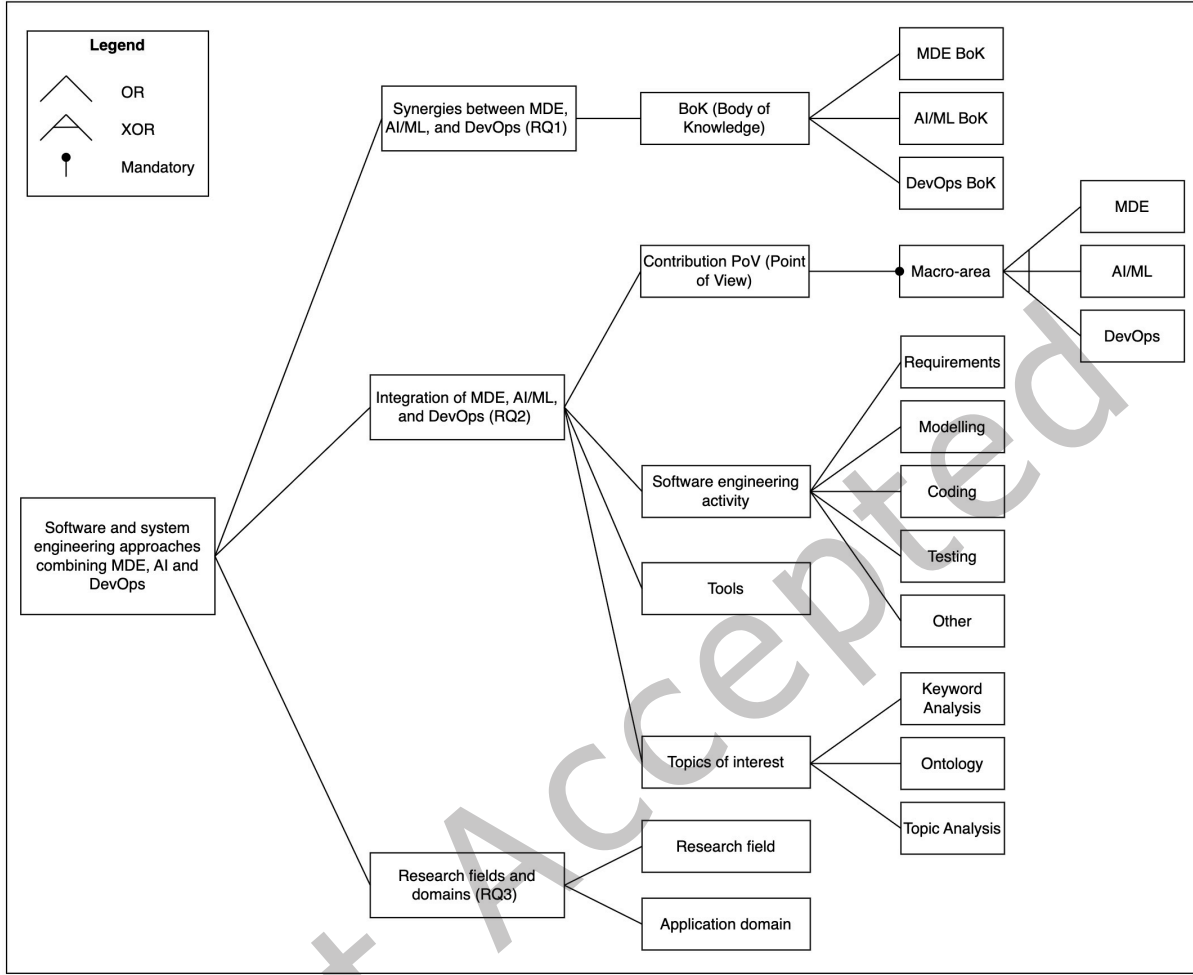


Fig. 2. Classification framework.

the nature of such synergies with a classification based on the Body of Knowledge (*BoK*) (described in Section 3.5.2).

- **Integration of MDE, AI/ML and DevOps (RQ2)**: Based on the paper review, we consider the main activities of *software engineering*, requirements engineering, modeling, coding, testing, monitoring and possibly others; We verify the availability of *tools* to support (software) engineering activities. Additionally, we examine the *topics of interest* using keywords, ontologies, and topic analysis methods (described in Sections 3.5.3, 3.5.4, and 3.5.5).
- **Research fields and domains (RQ3)**: Based on the information extracted from the paper review, we analyze the locations of the selected papers to identify the *research fields*. We identify the main application domains (automotive, aerospace, railway, maritime, construction, digital life, and manufacturing).

4 Results and Discussion

In this section, we describe in detail the results obtained from the full application of the SMS methodology previously presented in Section 3. We also analyze and discuss further these results to provide the reader with the main takeaways of our study.

4.1 Results overview

To start with, Figure 3 (a) gives an overview of the number of papers considered at each step of the search and selection process.

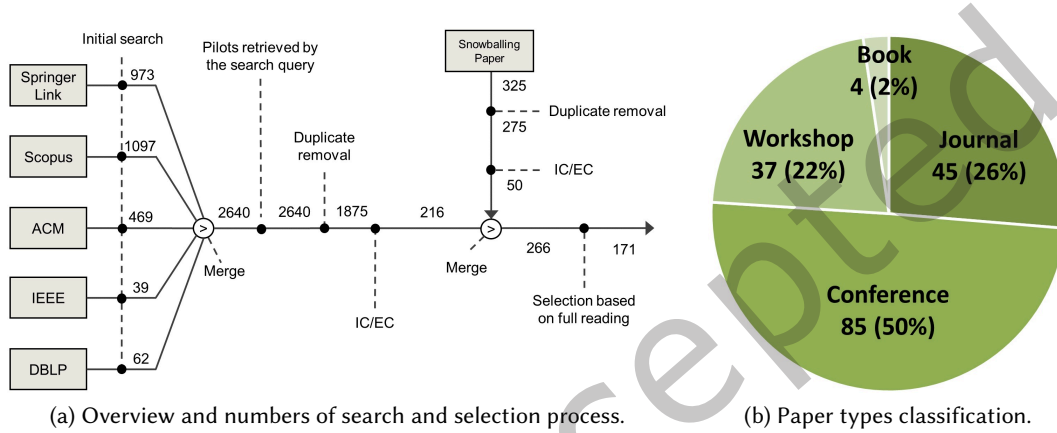


Fig. 3. Systematic mapping study results.

The number of published papers from the initial search, grouped by input database, is shown in Figure 3. The initial list, composed of 2640 papers, was reduced to 1875 after removing duplicates. Then, we applied the selection process (i.e., inclusion/exclusion criteria) on these 1875 papers to identify 216 papers possibly relevant to our study. After snowballing, this list has been extended up to 266 possible papers. Finally, all these 266 papers have been fully read to obtain the final list of 171 relevant published papers we considered in our study. This final list of selected papers (SP) is available in the Selected Paper additional material³.

4.2 Overall classification and publication trends

We first classified the selected papers according to their publication type: journal papers, conference papers, workshop papers, and book chapters¹¹. Figure 3 (b) shows the result of this classification. The majority of the selected papers have been published in conferences (85), journals (45), workshops (37), and book chapters (4). This can be considered coherent with the common practice of publishing main research results in relevant conferences within the Systems and Software Engineering community (contrary to other research communities where the priority is systematically given to journal publications).

We also looked at the publication trends over the years from 2009¹² up to January 2024. Figure 4 (a) distributes the 171 selected papers according to the years, starting from 2009. It shows an increasing interest, notably from 2015, in the synergy between MDE, DevOps, and AI/ML. The major boom is observed from 2019 to 2022, which coincides with the advent of DevOps and AI/ML in industrial and academic environments. The lower number

¹¹Whole books have not been considered, see exclusion criteria.

¹²The term DevOps is considered coined by Patrick Debois during the DevOpsDays event in Ghent: <http://www.devopsdays.org>.

of papers observed in 2023 may be due to the fact that the proceedings of many 2023 conferences (especially those taking place during the Fall) and some journal issues were not published yet or not uploaded to the various databases when the search queries were executed (i.e., in January 2024). However, we can reasonably expect a similar trend in 2023 (and 2024) than in the previous years. The decreasing trend for journals from 2021 to 2023 and the high number of papers published in workshops in 2019 and 2020, followed by an increase in conferences in 2021 and 2022, are closely related to improvements in work over the years, the need to cope with the fast-evolving pace of AI/ML, and the longer reviewing process of journals compared to workshops and conferences.

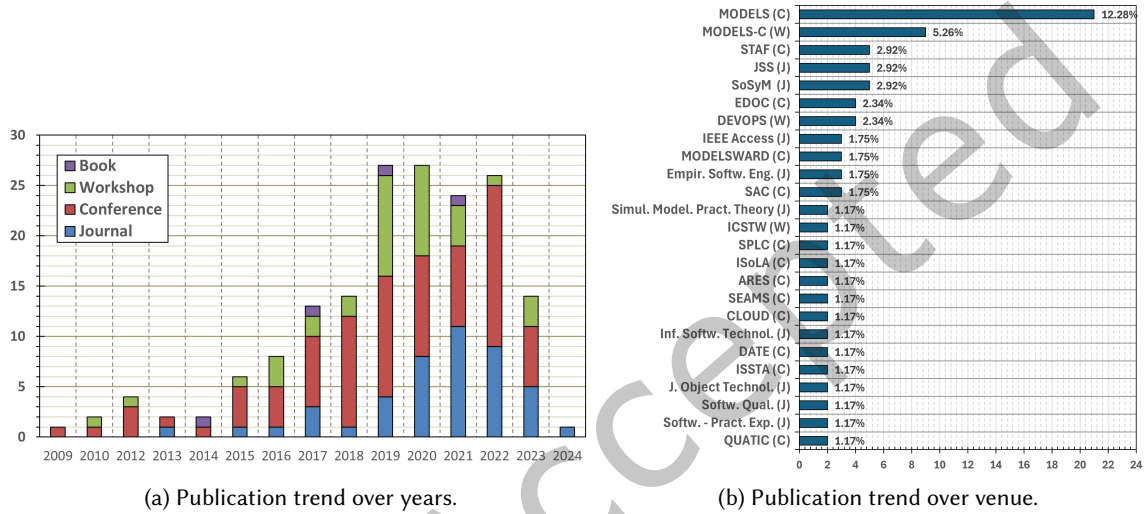


Fig. 4. Selected papers trends. Journal (J), Conference (C), Workshop (W), Book (B).

Figure 4 (b) shows the distribution of the selected papers across venues. The majority of the papers were presented at the MODELS conference or in MODELS-C workshops organized within the conference itself (30 papers, or almost 17.5% of the total number of papers). The SoSyM and JSS journals and the STAF conference have 5 papers, while the DEVOPS workshop and the EDOC conference have 4. The SAC and MODELSWARD conferences and the Empirical Software Engineering and IEEE Access journals follow with 3 papers each. All the other identified venues have only 2 and 1 selected paper. These numbers highlight the current major importance of DevOps and AI/ML within the Systems and Software Modeling community, which traditionally conducts research on MDE and its various applications to several domains.

Among all the selected papers, Table 6 details the 20 most cited ones according to Scopus and Scholar. From these 20 papers, 7 were published in journals, 10 in conferences, and 3 in workshops. Interestingly, the venues where these most cited papers have been published do not correspond to the most frequent venues we identified before (cf. Figure 4 (b)). This can be explained by the fact that these most cited papers mainly focus on DevOps and AI, while the most frequent venues we observed are rather MDE ones.

For example, the most cited paper [SP130] has been published at the *41st International Conference on Software Engineering: Software Engineering in Practice (ICSE-SEIP)* in 2019. It has 492 citations in Scopus and 860 in Google Scholar. The paper reports on a study conducted at Microsoft on real use cases by observing software teams developing AI-based applications for their portfolio of software products. As a result, they propose a nine-stage

Table 6. Selected papers with the most citations according to Scopus and Google Scholar (Jan. 2024). The last three columns represent the total number of citations by the other selected papers and the coupling papers (i.e., papers with the maximum number of overlapping references).

ID	Year	Source Title	Document Type	Scopus	Scholar	Citation by Selected Papers	Coupling Paper ID	Overlapping No. of References
[SP130]	2019	ICSE-SEIP	Conference	492	860	3	[SP69]	3
[SP119]	2017	ISSTA	Conference	163	269	6	[SP91]	6
[SP5]	2013	IEEE Trans. Softw. Eng.	Journal	87	147	1	[SP51]	5
[SP132]	2016	Future Gener. Comput. Syst.	Journal	66	109	5	[SP103]	6
[SP41]	2017	J. Syst. Archit.	Conference	64	104	0	[SP5]	2
[SP45]	2014	UCC	Journal	64	107	1	[SP46]	16
[SP46]	2018	ACM Trans. Internet Technol.	Journal	57	90	3	[SP45]	16
[SP64]	2015	MiSE	Workshop	54	77	1	[SP5]	1
[SP39]	2015	ISSTA	Conference	52	80	0	[SP8]	1
[SP88]	2020	ICSE	Conference	51	73	3	[SP58]	6
[SP29]	2018	ESEC/FSE	Conference	50	83	1	[SP73]	2
[SP55]	2017	DATE	Conference	49	73	0	[SP51]	1
[SP19]	2013	CBSE	Conference	43	60	4	[SP53]	7
[SP36]	2019	MODELS	Conference	36	39	2	[SP57]	16
[SP143]	2010	INDIN	Conference	36	44	0	N.A.	0
[SP47]	2017	STAF	Workshop	32	63	1	[SP78]	2
[SP100]	2016	QUDOS@ISSTA	Workshop	32	49	6	N.A.	0
[SP4]	2019	Softw. - Pract. Exp.	Journal	30	39	4	[SP109]	4
[SP62]	2020	IEEE Internet Things J.	Journal	26	38	0	[SP11]	1
[SP109]	2022	J. Syst. Softw.	Journal	25	33	2	[SP95]	8

workflow process informed by prior experiences developing AI applications. Thus, the paper can be classified as “DevOps for AI/ML” in our study.

Three papers [SP119][SP39][SP100] have been published at the *ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)* with 247 citations in total on Scopus and 398 on Google Scholar. In particular, one of these papers [SP119] introduces Retecs, a new method for automatically learning test case selection and prioritization in CI. The objective is to minimize the round-trip time between code commits and developer feedback on failed test cases. To this end, Retecs uses reinforcement learning to select and prioritize test cases according to duration, previous execution, and failure history. Thus, this paper can be classified as “AI/ML for DevOps” in our study. Another paper [SP39] contributes to finding performance bottlenecks of applications. It uses a genetic algorithm as a search heuristic of combinations of input parameter values that maximize a fitness function representing the elapsed execution time of a web application. Thus, this paper can be classified as “AI/ML for DevOps” in our study. The last paper of the three [SP100] introduces the DICER tool and provides an overview of how it can be used in a DevOps environment. DICER combines MDE tenets with the challenges of delivering a model-driven continuous deployment tool that allows quick (re-)deployment of data-intensive cloud applications. Thus, this paper can be classified as “MDE for DevOp” in our study.

Table 6 also presents a bibliographic coupling metric in the last two columns. The coupling metric measures the number of shared references between two documents, whereas the coupling paper is the paper with the maximum number of overlapping references (as visible in the last column). For instance, two papers [SP45] and [SP46] have 16 overlapping references. After double-checking, it appears that paper [SP46] is the follow-up of

paper [SP45]. The papers [SP36] and [SP57] also have 16 overlapping references. Thus, there is a clear connection between the two presented works.

4.3 Which approaches combining MDE, DevOps, and AI/ML have been proposed (RQ1)

Figure 5 (the green bars) shows that the 171 selected papers explicitly contributed to three main areas or *research areas*, which are MDE (146), DevOps (137), and AI/ML (75).

To answer RQ1, assessing whether and how such research areas have been intertwined is important. In this regard, we investigated the nature of such synergies among the three research areas. Since we obtained a limited number of papers that discuss all three dimensions together, we decided to include all synergies to help readers better understand the gaps in the literature and potential future directions, mapping the relevant papers and topics. In particular, we classified the selected papers in terms of active/passive roles within the possible synergies¹³. As a result, we distinguished six classes of papers according to the nature of their contributions, cf. the blue bars in Figure 5:

- **MDE for DevOps:** papers presenting model-driven approaches explicitly applied in a DevOps context (e.g., generating a textual CI/CD pipeline script from a model artefact via model-to-text transformation¹⁴);
- **MDE for AI/ML:** papers presenting model-driven approaches explicitly applied in AI/ML context (e.g., a metamodel/grammar specifying a domain-specific language for neural network specification);
- **DevOps for MDE:** papers presenting the application of DevOps principles and practices to support MDE approaches (e.g., a continuous integration of changes across different versions of models);
- **DevOps for AI/ML:** papers presenting the application of DevOps practices for AI/ML artefacts and processes (e.g., a CI/CD pipeline delivering AI/ML-augmented software components),
- **AI/ML for MDE:** papers presenting the application of AI/ML practices within MDE approaches (e.g., learning of actual parameter values replacing assumed ones in design-time models);
- **AI/ML for DevOps:** papers presenting the application of AI/ML practices in the context of DevOps artefacts and processes (e.g., updating steps of a CI/CD pipeline based on outcomes of ML algorithms applied to runtime data of the delivered application).

In addition, the orange bars in Figure 5 also provide a birds-eye view of research area suitability¹⁵. Overall, they suggest a stronger (mutual) integration between MDE and DevOps than between the other research area pairs (AI/ML and DevOps, MDE and AI/ML). Thus, the first and largest group of cross-fertilization contributions appears between MDE and DevOps (113). It relates firstly to applying MDE principles and practices to support DevOps (105) and secondly to adopting DevOps practices to support MDE (22). The second group, with 49 contributions, consists of MDE and AI/ML with balanced intertwining (25 MDE for AI/ML, 26 AI/ML for MDE). Finally, the third group of contributions is between DevOps and AI/ML (32 contributions), where the adoption of AI/ML principles and practices for DevOps concerns is predominant (11 vs. 22). Furthermore, as expected, the most challenging result is the one involving all areas (20 papers).

To complement the previously presented results, Figure 6 displays the synergies between the research areas (i.e., MDE, DevOps, and AI/ML) considered in the selected papers over the years. As can be observed, the majority of the papers refer to MDE for DevOps, with a concentration on publications from 2019 to 2022. We can also notice that there has been a notable increase in the number of papers referring to all the considered research areas during this same period. The increase in the use of AI/ML in DevOps and MDE is probably due to the

¹³It is important to note that changing the order of the terms completely alters the meaning. Specifically, we deal with AI/ML for DevOps (AIOps) and DevOps for AI/ML (MLOps). Furthermore, we intended to explore various synergies and directions (AI/ML for DevOps and vice versa) but not to go into detail on AIOps or MLOps as separate areas.

¹⁴The provided examples are deliberately left generic, thus not necessarily bound to any of the selected papers, to exemplify the influence of an *active* research area onto a *passive* one.

¹⁵Note that, if the value of the orange bar is less than the sum of the corresponding blue bars, this identifies overlapping contributions.

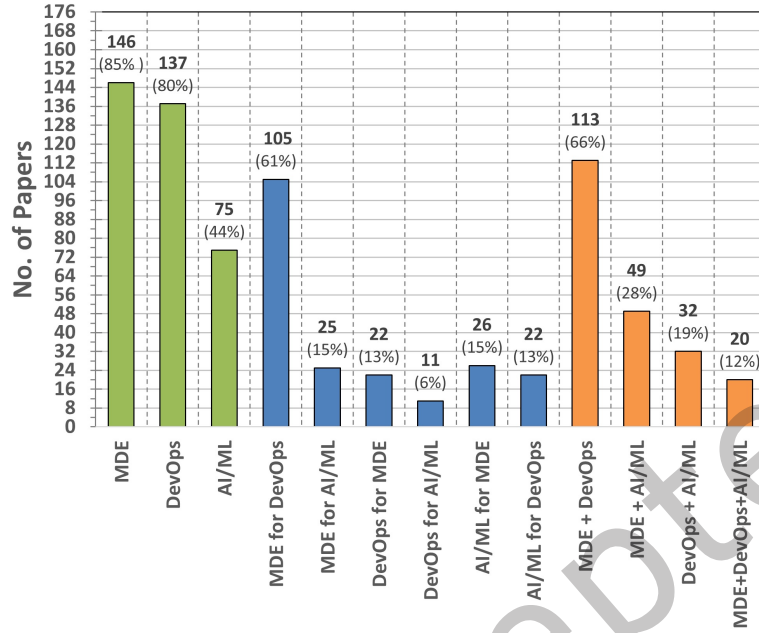


Fig. 5. Classification of the selected papers based on their contributions to MDE (a), DevOps (b), and AI/ML (c) research areas and their different synergies.

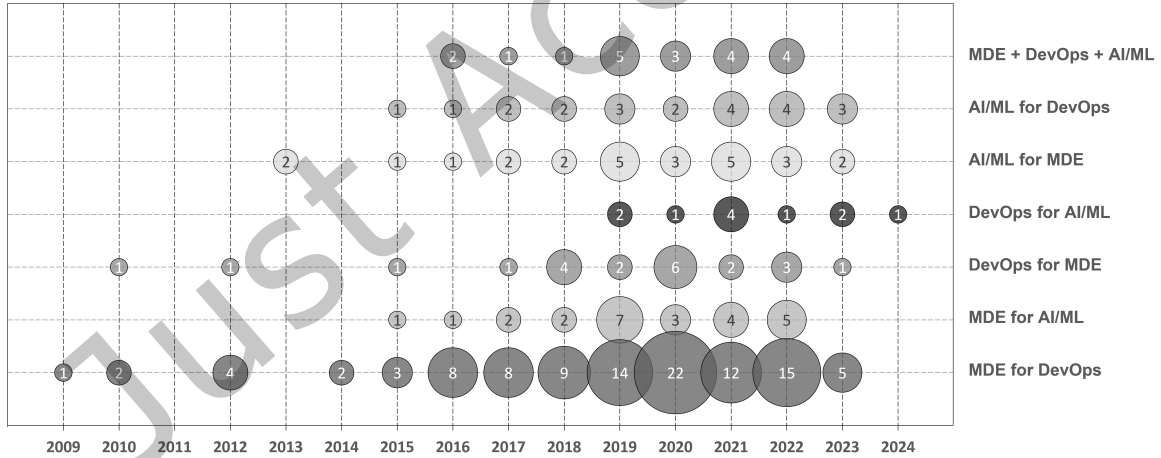


Fig. 6. Classification of the selected papers based on their combined contributions per Year (as of Jan. 2024).

advancement of ICT technologies that have allowed the use of AI/ML-based techniques to become widespread in various domains [32].

In the following sections, we present the classification of MDE, AI/ML, and DevOps (i.e., the synergies found in selected studies) based on their Body of Knowledge (BoK) (described in Section 3.5.2).

4.3.1 MDE and AI/ML Classification The Sankey diagram depicted in Figure 7 provides a quantitative representation of how research studies have addressed the MDE and AI intersection, highlighting the bidirectional influence between the two domains. The thickness of the connections between nodes corresponds to the number of studies discussing each synergy, offering insights into research trends and areas of focus. The interactive diagram is freely available online¹⁶.

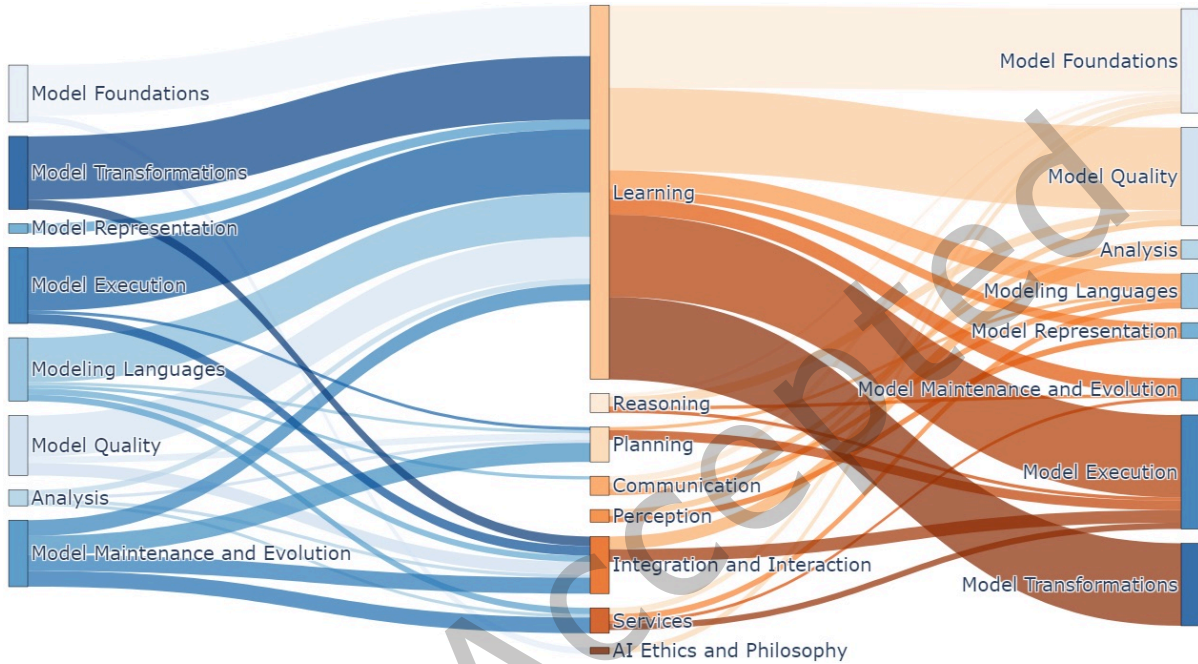


Fig. 7. MDE-AI classification based on their BoK. From left to right, the diagram represents the number of works dealing with MDE for AI (blue connection) and AI for MDE (orange connection), respectively.

A significant concentration of studies is observed in the **Learning** category (i.e., the ML category), with AI-driven techniques being extensively applied to MDE, particularly in **Model Foundations**, **Model Quality**, **Model Execution**, and **Model Transformations**. This suggests that ML is crucial in optimizing modeling processes, automating transformations, and improving model quality. Research such as [SP5] explores a learning-based framework for feature-oriented self-adaptive systems, showcasing how AI can enhance decision-making processes within MDE environments. Conversely, MDE has contributed to AI by providing structured approaches through metamodeling and domain-specific languages, as seen in [SP92], where frameworks like MontiAnna and ML-Quadrat support the structured development of ML models in IoT contexts.

Another notable observation is the relatively strong presence of research in **Integration and Interaction** and **Services**, where AI is leveraged to automate workflows, repair models, and enhance intelligent modeling assistance. Authors in [SP114] present a reinforcement learning-based approach to automated model repair, illustrating how AI techniques can dynamically improve model correctness and maintainability. Additionally, studies like [SP69] and [SP101] emphasize the role of automated model transformations in AI, demonstrating how MDE techniques can facilitate the seamless adaptation of AI architectures across different use cases.

¹⁶MDE-AI Classification: <https://hdl.handle.net/20.500.12004/1/U/167325>

The application of neural networks in intelligent model execution, explored in [SP105] and [SP140], further illustrates the potential of AI to enhance the runtime adaptability of models, allowing for dynamic decision-making and system optimization.

Despite the strong presence of research in **Learning** and **Integration and Interaction**, Figure 7 reveals gaps in **Reasoning**, **Communication**, and **Perception**, suggesting that logical inference, model-driven communication frameworks, and perception models remain underexplored. While there is extensive work in the validation and verification of AI systems through formal methods, as seen in [SP170] and [SP36], there is room for further investigation into how reasoning mechanisms can be better integrated into MDE environments. AI-driven model quality assessment, such as the approaches discussed in [SP115] and [SP79], showcases how data mining and deep learning techniques can be employed to evaluate the correctness and comprehensibility of models, yet the application of similar techniques to reasoning remains limited.

4.3.2 DevOps and AI/ML Classification Figure 8 illustrates the connections between the BoK of DevOps and AI, mapping key concepts and best practices that define their interplay. It highlights how AI techniques enhance DevOps and how DevOps methodologies streamline AI development. The interactive diagram is freely available online¹⁷.

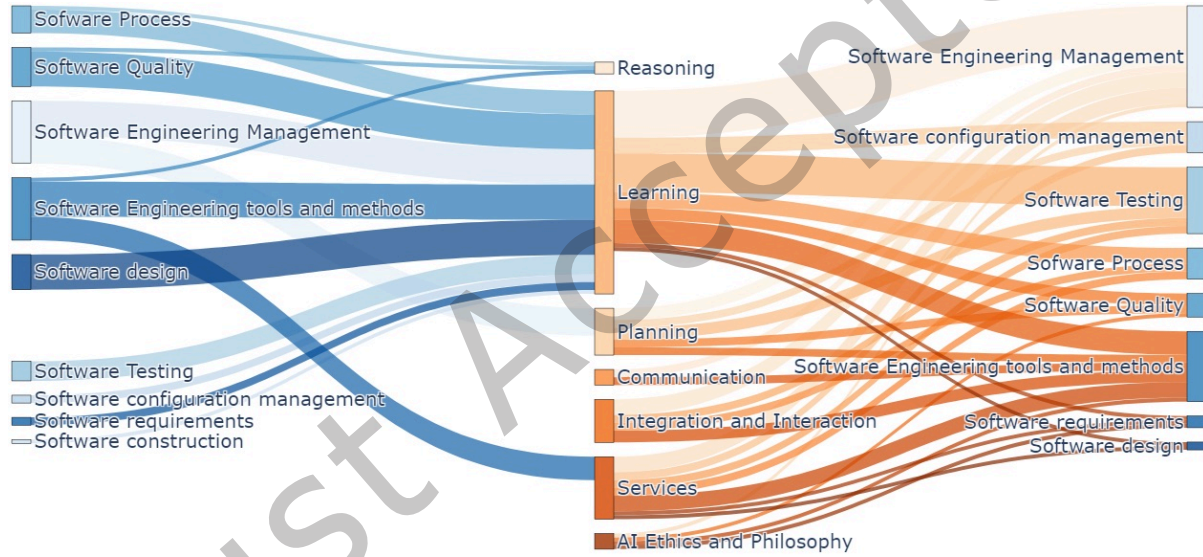


Fig. 8. DevOps-AI classification based on their BoK. From left to right, the diagram represents the number of works dealing with DevOps for AI (blue connection) and AI for DevOps (orange connection), respectively.

DevOps techniques for AI focus on automating and standardizing the ML lifecycle. Castellanos et al. [SP18] and Atouani et al. [SP32] apply DevOps principles to ML by integrating build automation and data management, while authors in [SP66, SP101] emphasize automated pipelines for model deployment and monitoring. Domain-specific applications, including industrial metal additive manufacturing and 6G network slicing [SP73, SP127], show how DevOps supports structured AI model deployment and performance optimization.

Conversely, AI techniques improve DevOps by enhancing testing, monitoring, and automation. Works in [SP2, SP15] use ML for test prioritization and regression testing, reducing costs while maintaining software

¹⁷DevOps-AI Classification: <https://hdl.handle.net/20.500.12004/1/U/359168>

reliability. Reinforcement learning strategies [SP23, SP119] optimize test execution dynamically, while predictive analytics [SP24, SP81] aid in failure detection and resource allocation. AI-driven security and compliance frameworks [SP169] automate vulnerability assessments, and transfer learning techniques [SP137, SP171] refine test prioritization and algorithm selection, making DevOps processes more adaptive.

Despite these advancements, Figure 8 also reveals notable gaps between DevOps and AI integration. Some areas, such as AI-driven software construction and perception, remain underexplored. The lack of reasoning-based AI methodologies in DevOps suggests an opportunity for enhancing intelligent automation through advanced decision-making frameworks. Furthermore, AI-driven communication strategies within DevOps remain an emerging field, indicating the need for more research into collaboration tools powered by natural language processing and knowledge graphs. Addressing these gaps could lead to more robust, adaptive, and intelligent DevOps pipelines.

4.3.3 MDE and DevOps Classification Figure 9 illustrates the interplay between MDE categories and DevOps practices, highlighting how different works bridge the two realms. The thickness of each flow indicates the number of papers dealing with the specific intersection of categories. For instance, one thick stream connects “Software construction” to “Model Transformations”, showing how multiple contributions (e.g., [SP54, SP78]) propose transformation pipelines for continuous integration. Conversely, a thin stream or no stream indicates fewer or no papers were found addressing that particular synergy. Although many contributions demonstrate a strong synergy between MDE and DevOps, some specific topics reveal limited coverage or are not explored, as is the case for “Software Quality” DevOps Practices. The interactive diagram is freely available online¹⁸.

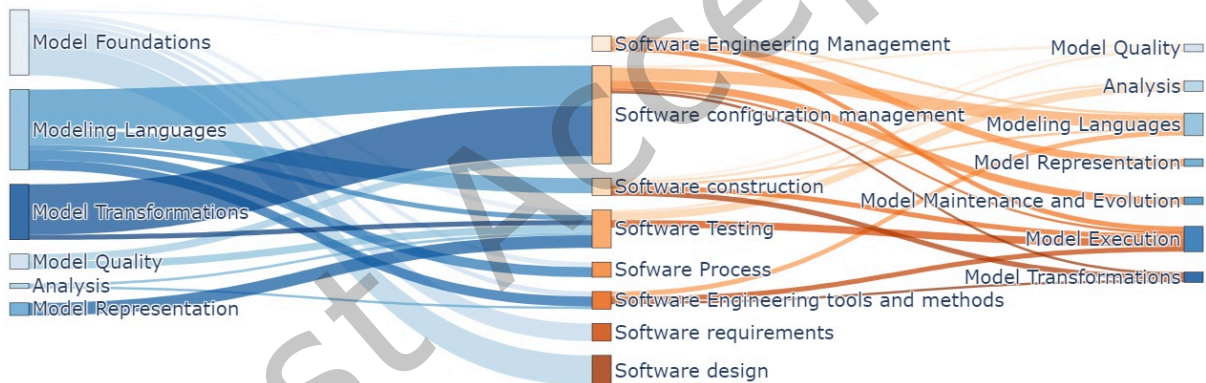


Fig. 9. MDE-DevOps classification based on their BoK. From left to right, the diagram represents the number of works dealing with MDE for DevOps (blue connection) and DevOps for MDE (orange connection), respectively.

Many of these works use domain-specific languages (DSLs) or other high-level modeling approaches to automate traditionally error-prone activities in DevOps pipelines, such as configuring continuous integration servers and automating deployment scripts. The authors in [SP53, SP109] exemplify this trend by showing how DSLs let teams abstract away complex IoT or edge deployments. In this manner, model transformations [SP54, SP78] reduce the manual effort of writing scripts and help unify processes across disciplines. Additionally, MDE improves automated testing in DevOps contexts, as illustrated by Wiechowski et al. [SP34], which proposes a DSL to generate test cases that integrate with continuous integration systems. The authors in [SP22] adopt

¹⁸MDE-DevOps Classification: <https://hdl.handle.net/20.500.12004/1/U/662594>

a model-based approach to regression testing, adjusting tests automatically in response to changing software behaviors.

Furthermore, DevOps principles enhance MDE processes by promoting shorter release cycles and greater automation. Authors in [SP140] show how continuous integration can quickly detect and correct issues in modeling artifacts, enabling more robust model quality checks. Some work leverages runtime data to inform and refine design-level models, as shown in [SP42], where operational metrics feed back into Model-Driven Architecture (MDA) decision-making. Other works apply continuous delivery not only to code but also to modeling infrastructures. For example, the authors in [SP89, SP131] show how meta-models can evolve quickly and safely through versioned pipelines and automated deployment. This approach ensures that teams adopting a model-driven approach can still benefit from the agility of modern DevOps environments, receiving real-time feedback about any changes in modeling artifacts. Concrete examples of these complementary strengths include the work in [SP63], which proposes explicit modeling of DevOps pipelines so that teams can automatically generate configurations for an entire toolchain. The authors in [SP167] report on an industrial experience where MDE is plugged into DevOps pipelines, allowing for fully automated builds of model-based artifacts. Similarly, Estivill-Castro et al. [SP107] integrate test-driven development at the model level, cutting down on time-consuming manual reviews. Finally, the authors in [SP99, SP164] show how DevOps best practices in change management, continuous adaptation, and scalability can be seamlessly integrated with MDE techniques for deriving new products and managing large models in low-code ecosystems.

Although there are notable synergies between DevOps and MDE, some areas have little or no attention. Certain MDE fundamentals, such as abstract syntax or formal semantics, rarely appear alongside DevOps, revealing a gap in fundamental modeling concepts in agile settings. Otherwise, DevOps aspects such as continuous monitoring, dashboards, or performance metrics are rarely combined with MDE approaches, pointing to untapped opportunities for using runtime monitoring data to refine design-level models. A similar gap appears in “*Software Quality*” for DevOps, particularly “*Use of data to guide quality assurance*”. While many studies combine MDE with testing or continuous integration, only a few show how large-scale operational data can improve models and quality assurance processes. Another underexplored theme is using real-time user or stakeholder feedback for requirements engineering in a DevOps context. Although some works hint at this, more research is needed to systematically feed that information into the early modeling phases.

4.3.4 MDE, DevOps, and AI/ML Synergies Finally, Table 7 highlights the selected 20 papers out of a larger set of 171, all of which address DevOps, MDE, and AI/ML in different software engineering contexts. Although relatively few, these contributions reveal current trends, practical applications, and potential gaps in how these three fields can be combined. The interactive Sankey diagram is freely available online¹⁹.

First, Castellanos et al. (2019) [SP93] start by combining MDE for model quality and transformations with DevOps testing and process management, using AI to study system performance under various conditions. Built on this work, Castellanos et al. (2020) [SP10] employ domain-specific models focused on model foundations, representation, and transformations to support Big Data Analytics in avionics through automated deployment, configuration management, and continuous performance monitoring while leveraging ML to automate workflows. Castellanos et al. (2021) [SP18] further advanced their approach by using similar models to set architectural inputs and deployment strategies, integrating DevOps automation and quality monitoring with AI-based learning algorithms. Langford et al. (2022) [SP13] introduce a modular, model-driven framework for managing learning-enabled components in safety-critical systems, merging MDE techniques for model representation and transformation with continuous integration and modular deployment from DevOps and deep learning for learning assurance. Van Den Heuvel et al. (2020) [SP101] adopt a model-driven MLOps approach, treating AI/ML blueprints as first-class models, leveraging MDE capabilities in modeling languages and transformations, aligning with DevOps

¹⁹MDE-AI-DevOps Classification: <https://hdl.handle.net/20.500.12004/1/U/325874>

Table 7. MDE-DevOps-AI/ML selected papers results. Aer. (Aerospace), Aut. (Automotive), D.L. (Digital Life), I. (Application Domain Independent).

Selected Paper	MDE→DevOps	MDE→AI/ML	DevOps→MDE	DevOps→AI/ML	AI/ML→MDE	AI/ML→DevOps	MDE Process	MDE Product	MDE Resource	DevOps Process	DevOps Product	DevOps Resource	AI/ML Process	AI/ML Product	AI/ML Resource	Requirements	Modelling	Coding	Testing	Monitoring	Domain(s)
[SP10]	■	■	-	-	-	-	■	-	-	■	-	-	-	■	-	■	■	-	-	■	Aer., I.
[SP13]	■	■	-	-	-	-	■	-	■	-	-	-	-	-	-	-	-	-	-	-	Aut., I.
[SP18]	■	-	-	■	-	-	-	-	■	-	-	-	-	-	-	-	■	-	-	-	Aer.
[SP20]	■	■	-	-	-	-	■	■	-	■	-	-	-	-	-	-	■	■	■	-	Aut.
[SP24]	■	-	-	-	-	■	■	-	-	■	-	-	■	-	■	-	-	-	-	-	I.
[SP29]	■	-	-	-	-	■	-	-	■	■	-	-	■	-	-	-	-	-	■	-	I.
[SP32]	■	■	-	■	-	-	■	-	■	■	-	-	-	■	-	-	-	-	-	-	I.
[SP40]	■	■	-	-	-	-	■	-	-	-	■	-	-	-	-	■	■	-	-	-	I.
[SP50]	■	-	-	-	■	■	-	-	■	■	-	-	-	-	■	-	-	-	■	■	I.
[SP52]	■	■	-	-	-	-	■	-	-	■	-	-	■	-	-	-	■	-	-	-	I.
[SP67]	■	■	-	-	-	■	-	-	■	■	-	-	■	-	-	-	-	-	-	-	I.
[SP84]	■	-	-	-	■	-	-	■	-	■	-	-	-	■	-	■	-	-	-	-	I.
[SP93]	■	■	-	■	-	-	■	■	■	■	-	-	-	-	-	■	■	-	-	■	Aer., D.L., I.
[SP94]	■	-	-	-	■	-	-	■	-	-	-	-	-	■	-	■	■	-	-	-	I.
[SP98]	■	■	-	-	-	-	■	-	-	■	-	-	■	-	-	■	■	-	-	-	I.
[SP101]	■	■	-	■	-	-	■	■	■	■	-	-	■	-	-	■	■	■	■	■	D.L., I.
[SP142]	■	-	-	-	■	-	■	-	-	-	-	-	■	-	-	-	-	-	■	-	I.
[SP161]	■	-	-	-	■	-	-	■	■	-	-	-	-	-	-	-	-	-	-	-	I.
[SP170]	■	■	-	-	-	-	■	-	-	■	-	-	■	-	-	-	■	-	-	■	I.
[SP171]	-	■	-	-	-	■	■	-	■	■	-	-	■	■	-	-	■	■	-	-	I.

strategies for continuous deployment, testing, and monitoring, and incorporating AI techniques such as learning, reasoning, and planning. Hu et al. (2018) [SP29] synthesize robust, reusable UI tests by employing model-based representations of UI flows, automating test generation and continuous testing, and using AI for perception and learning tasks. Atouani et al. (2021) [SP32] support deep learning frameworks through artifact and reference models, demonstrating MDE's metamodeling, analysis, and transformation capabilities, combined with DevOps practices like automated build systems, data management, and deep learning algorithms. Song et al. (2020) [SP98] present a model-based approach for fleet deployment in edge computing applications using platform-independent models and constraint solving, integrated with automated deployment, configuration management, and AI-based context-aware decision-making. Baumann et al. (2022) [SP67] address dynamic data management and the continuous retraining of ML models with a model-driven strategy that structures iterative data management and supports continuous integration and retraining cycles, emphasizing AI's learning components. Kantel et al. (2019) [SP142] offer a scalable testing toolkit for autonomous systems using formal models, such as timed automata, to structure tests within continuous testing pipelines, also incorporating AI for verifying autonomous behaviors. Lapointe-Boisvert et al. (2021) [SP161] explore the modeling of acceptance tests to facilitate automated software measurement, leveraging MDE for test representation and aligning with DevOps continuous feedback loops, while suggesting potential AI techniques for analyzing testing artifacts. Langford et al. (2022) [SP13] devise a modular framework to boost AI reliability through loosely coupled services, while Rosch et al. (2022) [SP20] tackle the integration of ML models into automotive architectures, and Kampik et al. (2019) [SP24] blend agents and

business processes to manage IoT systems. Hahnle et al. (2018) [SP50] focus on maintaining software consistency through continuous checks, and Bersani et al. (2016) [SP52] provide a tool called *OSTIA* to identify anti-patterns in stream-based big data pipelines. Bergelin et al. (2022) [SP84] synthesize industrial requirements for AI-driven MDE, Liu et al. (2021) [SP94] propose a maturity metric for composing model services in cloud environments, while Bersani et al. (2019) [SP170] extend the *OSTIA* tool to verify big data topologies. Finally, Benni et al. (2019) [SP171] propose a meta-model for managing ML pipelines that exemplifies the integration of MDE through model transformations and meta-modeling, ensures seamless continuity between development and operations via DevOps practices, and incorporates AI meta-learning techniques to optimize pipeline selection and performance.

Although this work has succeeded in extracting several fields explored across these papers, it is clear that the identified gaps are numerous. In terms of MDE, there is a lack of end-to-end model-driven approaches covering the full lifecycle in AI-driven systems (e.g., from requirements and design to runtime adaptation and continuous monitoring), as highlighted by Naveed et al. [46]. Concerning AI, the integration of ethical and philosophical considerations, as well as communication, integration, and interaction aspects, is not well addressed, and advanced reasoning and planning techniques are not deeply embedded within DevOps decision-making processes. From the DevOps perspective, while automation and continuous integration are well represented, there remains a shortfall in integrating AI-driven meta-learning to inform and optimize model-driven transformations and deployment pipelines.

4.3.5 RQ1 Discussion. Globally, there is a growing trend in the publication of conference papers over the years, especially in software and systems engineering conferences. This is primarily due to technological advancements, which have led to the increasingly pervasive use of AI/ML techniques and the high speed at which papers are written and published in this particular domain. Authors tend to favour conferences with shorter response times to publish their work as quickly as possible. Publishing longer papers presenting more extended works in journals only comes as a second step. Software and systems engineering journals are the most prominent among those we analyzed in this SMS. On the contrary, relatively few publications are coming from journals or conferences focusing particularly on AI/ML aspects. This is possible since many research efforts involving the use of AI/ML are instead published in venues specific to given application domains or areas of interest.

As introduced before, our search considers each of the 3 research areas covered by the study; the used query does not exclude any of them. This means that when we discuss pairs of research areas, we discuss contributions that fall within the complete topic but explicitly implement only 2 research areas. This premise leads us to state that there is a predominant attention toward using MDE techniques to support the continuous development and DevOps process.

This confirms the relevance of model-based approaches for DevOps as a novel research direction studying the automated connection of CPS design models with CPS runtime monitoring in the form of Digital Twins [17]. All the other synergies are significantly lower, suggesting that many challenges involving DevOps and AI/ML combined with model-based and model-driven technologies are still open.

Specifically, the overall trend in MDE and AI suggests that learning-based approaches dominate the intersection of MDE and AI, strongly emphasizing improving model transformations, validation, and execution. Reinforcement learning, automated reasoning, and neural networks have proven to be valuable assets in enhancing model adaptability and quality. At the same time, MDE techniques such as metamodeling and DSLs provide the structural foundation for AI development. The integration of these methodologies has led to significant advancements in AI-powered MDE ecosystems, yet challenges remain in areas such as automated reasoning, perception-driven modeling, and communication frameworks. Future research should aim to bridge these gaps by developing novel AI-driven techniques tailored to enhance reasoning capabilities in MDE, while also leveraging MDE principles to further structure and optimize AI-driven systems. More robust, scalable, and intelligent MDE frameworks can

be realized through the continued convergence of these fields, fostering greater automation, adaptability, and efficiency in both AI and MDE domains.

On the other hand, as DevOps and AI continue to evolve, automation and intelligence become central to software engineering. Trends indicate increased AI-driven automation in DevOps, fostering self-healing systems and reducing manual intervention. The expansion of MLOps frameworks, reinforcement learning in CI/CD, and AI-driven security solutions are shaping the future of software automation, improving efficiency, transparency, and scalability.

Furthermore, there is significant evidence that bringing MDE techniques, such as DSLs, model transformations, and automated analysis, into DevOps pipelines can streamline and automate complex tasks, especially in testing and deployment. Conversely, DevOps practices help make MDE processes faster, more iterative, and more attuned to stakeholder requirements. Nevertheless, our work also reveals missing links, including little attention to foundational MDE concepts in DevOps contexts and limited exploration of continuous monitoring or data-driven quality assurance in MDE software lifecycles. Filling these gaps could expand the current state of the art, leading to an even tighter and more robust integration of MDE and DevOps principles for developing, operating, and evolving modern software systems.

Finally, the works in Table 7 demonstrate that MDE can structure and automate software processes, DevOps enables continuous and iterative integration, and AI/ML enriches systems with data-driven intelligence. Nevertheless, despite these promising avenues, the limited number of papers addressing all three dimensions highlights important gaps: i) there is still a need for more systematic integration of MDE artifacts and DevOps pipelines tailored specifically to ML lifecycles, and ii) a broader exploration of automated testing and monitoring for ML-driven architectures. Moreover, it is worth noting that MDE appears to dominate DevOps and AI, mainly due to the essential abstraction engineering activities necessary to set up the entire process. In this respect, there seems to be a lack of a more holistic discipline aimed at continuously managing abstraction, including data representation.

4.4 Which software engineering activities and perspectives have been considered (RQ2)

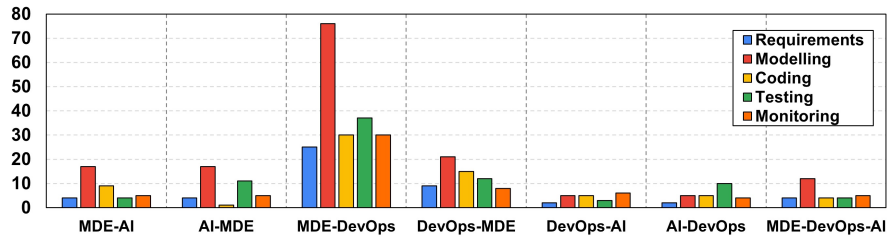
To answer RQ2, we analyzed several aspects, as described in the following.

4.4.1 Software engineering use case scenarios Following the data extraction process presented in Section 3, we considered five main software engineering activities or use case scenarios (related to DevOps): *requirements engineering*, *modelling*, *coding*, *testing*, and *monitoring*. In our study, a selected paper was related to a use case scenario when at least one *process*, *resource*, or *product* is presented in the paper to support such a use case scenario. It is worth noting that a single selected paper can support multiple use case scenarios in various ways.

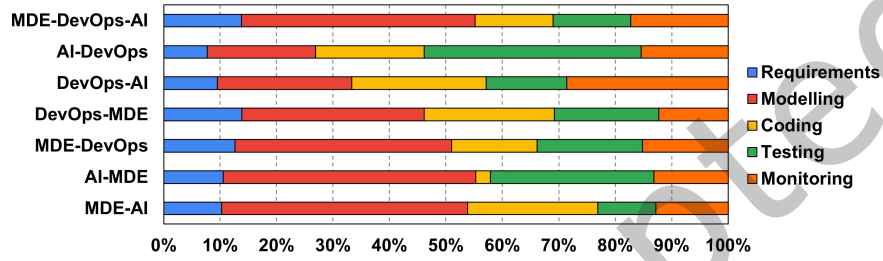
The most common scenario is modeling (111), followed by testing (64) and coding (48). Our study shows that monitoring (41) and requirements (36) have received slightly less attention.

Furthermore, Figure 10 (a) and Figure 10 (b) provide a visual representation of how different software engineering activities are addressed within the selected papers.

Figure 10 (a) shows an absolute distribution of the number of papers associated with each category. Modeling emerges as the dominant activity, with a particularly high presence in MDE-DevOps works. This is expected, given that MDE heavily relies on model creation and transformation as core practices. Testing and coding also show significant representation, particularly in MDE-DevOps and DevOps-MDE synergies, indicating that these areas are well integrated into modern SE workflows. Monitoring, while present, appears less frequently, suggesting that it remains an underexplored area in the context of MDE-DevOps-AI/ML integration, as presented in the previous sections. Requirements engineering is the least represented activity, highlighting a gap in research concerning systematic requirements specification using MDE and AI-driven DevOps.



(a) Absolute distributions.



(b) Percentage distributions.

Fig. 10. Software engineering distributions among the selected papers and dimensions.

Figure 10 (b) offers a normalized perspective by showing the relative proportion of each software engineering activity across different categories. It confirms the predominance of modeling but also sheds light on the differences in activity distribution. The presence of modeling remains strong in synergies involving AI, such as AI-DevOps and AI-MDE. Still, coding, testing, and monitoring are more evenly distributed, indicating that AI/ML applications are leveraged across multiple development phases. Conversely, MDE-DevOps scenarios exhibit a strong bias toward modeling, coding, and testing, with monitoring and requirements engineering receiving comparatively less attention.

These results suggest that while modeling is a well-established aspect of MDE, its integration into DevOps and AI-driven workflows still faces challenges, particularly in ensuring continuous requirements engineering and monitoring, as stated in the previous sections. The limited focus on monitoring is particularly critical, as modern DevOps pipelines emphasize observability, feedback loops, and data-driven decision-making, aspects that AI/ML techniques could enhance. The relatively low emphasis on requirements engineering also suggests that the potential benefits of model-based requirements specification and automated traceability have not yet been fully explored in the context of MDE-DevOps-AI/ML. Overall, Figure 10 reinforces the idea that research at the intersection of MDE, DevOps, and AI/ML tends to prioritize modeling and testing over other activities. This highlights opportunities for future research to strengthen the role of requirements engineering and monitoring, potentially leveraging AI-driven approaches to automate and enhance these phases within DevOps pipelines.

In a long-term practical study conducted within the context of five large European collaborative projects [55], Model-based Requirements Engineering (MBRE) has proven to bring benefits in terms of scalability (e.g., a large number of handled requirements), heterogeneity (e.g., partners with different types of RE background), adaptability and extensibility (e.g., to various project's needs), traceability (e.g., from the requirements to the

software components), automation (e.g., documentation generation), consistency and quality (e.g., central model), and usefulness or usability (e.g., actual deployment and practical use).

Beyond these statistical insights, the complete list of papers classified under each of the above-mentioned use case scenarios is an important outcome. Thanks to it, researchers and practitioners can explore the relevant literature to improve their work in the context of a specific scenario or activity.

4.4.2 Topics exploration This section outlines the results of the systematic approach to examining topics of interest using keywords, semantics/ontology, and NLP analysis methods.

Keyword Analysis: We extracted 724 keywords from all the selected papers. After cleaning the list of keywords, we ended up with 416 unique keywords. Some of the keywords occurred more often in the selected papers. For instance, 35 prevalent keywords occurred 4 to 37 times as shown in Figure 11²⁰. However, the large majority of the identified keywords only occurred once (332), twice (37), and three times (11) in the selected papers.

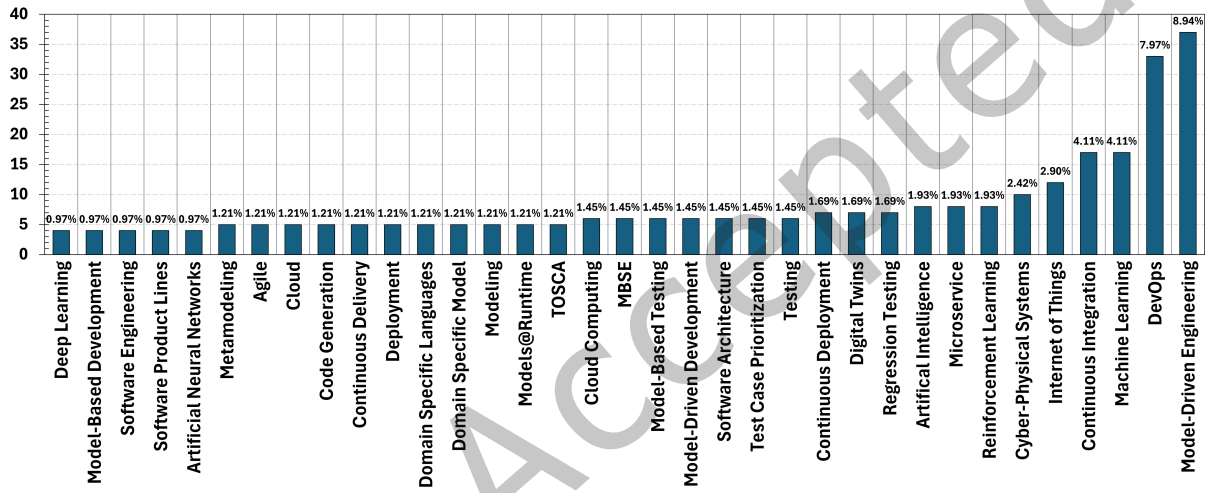


Fig. 11. Most Recurrent Keywords.

Ontology: As stated in Section 3.5.4, we have constructed an ontology encompassing the semantic relations between the keywords. The final ontology tree can be freely downloaded³. The ontology Classes/Slots also include frequency information (i.e., weighted relations) among the tree nodes using the *Weighted Keyword* (WK) metric defined in Section 3.5.4. Some examples are listed below:

- “Systems and Software Engineering” Class (WK = 293) encompasses a larger category due to the high number of keywords related to it;
- “Testing” (WK = 32) is included as a Class to group related concepts like the “Regression Testing” Class (WK = 10) and “Test Case Prioritization” Slot (WK = 6).
- “Machine Learning” Class (WK = 43) is the parent of “Deep Learning” Class with WK = 8 and “Artificial Neural Networks” Slot with WK = 4.

We conducted a further analysis to extract the relationships between the various Classes/Slots. We considered keywords related to “Model-Driven Engineering”, “DevOps”, “Artificial Intelligence”, and “Machine Learning”

²⁰Note that the term “TOSCA” stands for “Topology and Orchestration Specification for Cloud Applications” which is an OASIS standard used in several of our selected papers e.g., [SP132, SP102, SP103].

and checked for other keywords that appeared with them in the keyword list of each selected paper and belonged to another branch of the ontology (e.g., in [SP90] the keyword “DevOps” appears in the keyword list together with “Support vector machine” and “Artificial neural network”, so there is a relationship between the “DevOps” Class and the Slots “Support vector machine” and “Artificial neural network”, which belong to the path of the “Artificial Intelligence” Class in our ontology). From this analysis, we observe that:

- **MDE** is related to 1) “CPS”, particularly for “IoT” [SP141][SP159]; 2) “Software Engineering Activities” for “Requirements” [SP28][SP158] and “Coding” [SP25][SP76]; 3) “DevOps”, specifically in “Service-Oriented” [SP17] and “Microservices” [SP7][SP116] (i.e., architectural reusability and network services); 4) “ML” for “NLP” [SP110] [SP169] and “Recommendation Systems” [SP120].
- **DevOps** is related to 1) “CPS” for “IoT” [SP53][SP159] and “Digital Twins” [SP144][SP157], particularly in the “Automotive” domain [SP11][SP41]; 2) “Coding” [SP35][SP116] and “Performance” [SP7][SP93] for the “Software Engineering Activities” Class; 3) “Metamodeling” [SP147] and “Domain-Specific Activities” [SP93] for “MDE” Class; 4) Machine learning algorithms for “Regression” [SP4] and “Data Analytics” [SP10] for the “AI/ML” Class.
- **AI-ML** is related to 1) “CPS” for “Digital Twins” [SP121][SP166]; 2) “Software Engineering Activities”, particularly for “Testing” Class [SP29][SP119]; 3) “Metamodeling” [SP32][SP36] and “Domain-Specific Activities” [SP10][SP105] for “MDE” Class; 4) “Continuous Integration” [SP15][SP88] and “Machine Learning Operations” (MLOps) [SP121] [SP127] for “DevOps” Class. 5) “Autonomic Computing” [SP5] and “RLOps” (Reinforcement Learning Aided Open-RAN) [SP121] for the “Cloud” Class; 6) “Cybersecurity” [SP169] for the “Security” Class.

Additionally, MDE and DevOps research areas use tools based on Eclipse Modeling Framework, Sirius, and Ant build system [SP28][SP167], or JetBrains Meta Programming System (MPS) [SP155]. Standards such as TOSCA [SP103][SP132] and Open-RAN [SP121] are used for the “Cloud” Class, while AI/ML and DevOps use “Explanation-guided Reinforcement Learning” (XRL) for eXplainable AI (XAI) [SP127].

Topic Analysis: The result of the topic analysis process is presented in Table 8. The rank of the top 25 terms, which gives us a very raw understanding of the content of the selected paper collection, is as follows:

1. model (22.19), 2. software (18.99), 3. learning (18.85), 4. based (16.87),
5. devops (16.40), 6. systems (16.18), 7. testing (15.09), 8. engineering (14.77),
9. data (14.60), 10. models (14.38), 11. development (14.13), 12. continuous (13.08),
13. deployment (12.43), 14. driven (12.02), 15. integration (11.89),
16. approach (11.56), 17. cloud (11.26), 18. modeling (11.08), 19. applications (10.21),
20. process (10.09), 21. code (9.35), 22. requirements (9.09), 23. machine (9.08),
24. design (8.59), 25. architecture (8.49).

As stated in Section 3.5.5, a critical step involves selecting the number of topics “ k ”. We explored a wide range of k values [$k_{min} = 4, \dots, k_{max} = 40$] and evaluated topic coherence using the mean TC-W2V measure [47]. Since the k value that yielded the highest mean topic coherence is 38, we fixed this number and extracted 38 topics while connecting them to the ontology Classes/Slots. The highest-ranked topic (i.e., column Rank) is related to *Model-based Approaches* (Rank 1, [SP44]), followed by *Machine Learning Foundations* (Rank 2, [SP21]) and *Testing* (Rank 3, [SP16]). *Cloud Services* (Rank 4, [SP124]) is also an important topic of interest in this analysis, followed by *DevOps Practices* (Rank 5, [SP90]) and *Data-Driven Techniques* (Rank 6, [SP170]). *Continuous Deployment* (Rank 7, [SP134]) and *DevOps in Cyber-Physical Systems* (Rank 8, [SP75]), *Software Processes* (Rank 9, [SP50]) and *Requirement Engineering* (Rank 10, [SP110]), close this high-ranked list. From this topic analysis, it is clear that MDE, AI/ML, DevOps, CPSs, and Cloud domains, along with software and systems engineering approaches, play a fundamental role, leading to a paper classification based on the relative importance extracted from the analysis shown in Table 8.

It is worth noting that the identified topics are inevitably influenced by the underlying structure of the data corpus, and the automated topic analysis might not always produce insights as deep as those from manual sorting.

Table 8. Topic Analysis on Titles, Abstracts and Keywords

Ontology	Topics	Top-Term Weight	Rank	5 Most Relevant Papers Ranked by Importance
Systems and Software Engineering	Complex Systems Support	1.9915	38	[SP108][SP112][SP53][SP148][SP70]
	Software Processes	2.1594	9	[SP50][SP122][SP130][SP5][SP74]
	Engineering Processes	2.4045	12	[SP165][SP37][SP99][SP112][SP86]
Model-Based SW Engineering	Model-Based Approaches	3.4500	1	[SP44][SP120][SP23][SP2][SP25]
	Modeling	2.1372	24	[SP68][SP82][SP30][SP107][SP162]
Model-Driven Engineering	Model-Driven Techniques	2.4670	31	[SP89][SP47][SP114][SP87][SP162]
	MDE and Agile	2.1402	35	[SP167][SP136][SP92][SP154][SP63]
	Domain-Specific Tools	1.6586	25	[SP155][SP59][SP116][SP150][SP147]
	Modeling Tools	1.5309	29	[SP102][SP69][SP85][SP111][SP168]
DevOps	DevOps Practices	2.1182	5	[SP90][SP154][SP63][SP102][SP156]
	Operational Management	1.8959	19	[SP48][SP42][SP164][SP24][SP69]
	Continuous Development	2.7951	15	[SP12][SP144][SP48][SP78][SP116]
	Continuous Deployment	1.9364	7	[SP134][SP103][SP129][SP95][SP125]
	Automation Challenges	1.8224	21	[SP133][SP129][SP132][SP143][SP61]
	DevOps Tools	1.3957	22	[SP4][SP95][SP78][SP88][SP62]
	Integration	2.1960	23	[SP113][SP4][SP146][SP78][SP40]
Artificial Intelligence	Data-Driven Techniques	2.3225	6	[SP170][SP66][SP52][SP3][SP67]
	Machine Learning Foundations	2.1100	2	[SP21][SP22][SP139][SP87][SP119]
	Machine Learning Techniques	1.8138	33	[SP67][SP32][SP69][SP81][SP1]
Cyber-Physical Systems	DevOps in CPSs	1.7415	8	[SP75][SP6][SP140][SP136][SP151]
	Industrial Use Cases	1.0948	36	[SP65][SP56][SP135][SP84][SP138]
Cloud	Cloud Services	2.2855	4	[SP124][SP45][SP46][SP8][SP83]
Systems and Software Engineering Activities	Requirements Engineering	1.6205	10	[SP110][SP118][SP123][SP28][SP84]
	Modeling	1.8632	11	[SP152][SP54][SP104][SP106][SP149]
	Coding	1.5897	16	[SP60][SP76][SP38][SP44][SP164]
	Testing	2.0621	3	[SP16][SP142][SP29][SP15][SP2]
	Monitoring	1.3189	20	[SP166][SP62][SP85][SP50][SP55]
Others	Optimization Processes	1.2636	37	[SP24][SP20][SP38][SP14][SP106]
	Performance	1.4146	30	[SP39][SP93][SP7][SP18][SP51]
	Analysis Tools	1.9700	28	[SP72][SP80][SP134][SP87][SP50]
	Design Approaches	2.3148	26	[SP108][SP157][SP10][SP117][SP49]
	Architecture	2.1719	13	[SP31][SP51][SP41][SP118][SP24]
	Applications	2.1337	27	[SP109][SP149][SP135][SP101][SP43]
	Execution Time	1.8692	14	[SP43][SP39][SP103][SP98][SP101]
	Context-Driven Support	1.0001	34	[SP161][SP171][SP162][SP88][SP160]
	Quality Control	1.8809	17	[SP33][SP26][SP1][SP64][SP66]
	Frameworks	1.5753	18	[SP13][SP163][SP11][SP33][SP41]
	Components	1.8138	32	[SP145][SP49][SP35][SP98][SP58]

However, in large-scale secondary studies, manual classification becomes challenging. We adopted a mixed approach to address this, combining topic modeling with manual refinements. Topic modeling was primarily used to uncover insights that the authors may have overlooked, rather than serving as the main approach of the study. Therefore, in our study, our method provides a feasible balance between automated and manual analysis, as the topic analysis serves as an exploratory technique to guide further interpretations by the authors.

4.4.3 Tool Availability This section presents the availability of tools identified in our analysis that support several (software) engineering activities. In particular, we can highlight the following:

- 73 selected papers ($\approx 43\%$ of the total selected papers) provide means to obtain the tools mentioned in the papers. In comparison, 49 selected papers ($\approx 29\%$ of the total selected papers) do not provide clear information about tool availability. Indeed, the latter does not necessarily imply that tools are unavailable, but rather that this particular information is missing for different reasons (e.g., tool support was not in the scope of a given paper). The remaining studies do not report any developed tools but focus on methodological or conceptual research.
- Out of the 73 selected papers that referenced tool availability, 32 papers ($\approx 44\%$) have tools that are either not freely accessible or their links are not provided; 7 papers ($\approx 10\%$) offer outdated or non-existent links, while 28 papers ($\approx 38\%$) direct to links on GitHub. Additionally, 6 papers ($\approx 8\%$) provide links to their official project, tool, or organization pages. Moreover, only 34 papers out of 73 (i.e., $\approx 47\%$) provide access links to the tool or experimental results. In comparison, the remaining 39 papers (i.e., $\approx 53\%$) do not offer free access to any code, repository, or website²¹.
- Finally, we are not discriminating between open/closed source tools or free/commercial ones. Nevertheless, it is worth recalling that we excluded non-scientific publications and grey literature (cf. Table 3). Thus, we may expect some commercial tools to be excluded from the assessment of tool availability.

Table 9 offers an integrated overview of the 34 available tools implemented in the selected papers that combine MDE with AI/ML techniques and DevOps practices to support the full software engineering lifecycle.

Among the contributions, a significant group of tools focuses on MDE to bridge the gap between abstract models and executable systems. Tools such as JTL and ACCORDANT illustrate how MDE techniques enable continuous performance engineering in microservice-based systems and support architectural design in big data analytics applications, respectively. iObserve and CloudML leverage MDE to integrate runtime observations with development models and manage multi-cloud deployments using DSLs and models at run-time. Other MDE solutions (e.g., OSTIA, DICER, MontiThings, openCAESAR, Scalable Views) address challenges in architecture visualization, continuous deployment, reliable IoT application development, microservice integration, and the construction of scalable views on heterogeneous models. In contrast, some works adopt alternative approaches that do not primarily rely on MDE. For instance, Joda-Time offers a tag-based recommender system for regression test case prioritization, while DockerConflict statically analyzes and automatically rewrites container deployment artifacts.

A particularly notable subset of the MDE-based contributions further integrates AI/ML techniques with DevOps practices, resulting in enhanced automation, agility, and quality in software development. In this integrated approach, ACCORDANT leverages model-driven techniques for the architectural design of big data analytics applications, enabling the automated specification and monitoring of quality attributes. OSTIA employs a model-driven methodology to continuously analyze and refine stream-based architectures, automatically detecting design anti-patterns and supporting runtime verification. Finally, DivENACT demonstrates the power of model-based automation in deploying edge computing applications by extracting platform-independent models and using constraint solving to assign deployments to heterogeneous device fleets.

Furthermore, the distribution of application domains among the tools reveals that only a few have been validated in specific domains compared to the larger number of domain-independent approaches. Tools such as ACCORDANT and openCAESAR are designed to tackle the challenges of complex, high-reliability aerospace applications. In the automotive sector, FFSM Diff stands out as a dedicated tool for behavioral modeling in software product lines. It utilizes adaptive learning techniques to reuse behaviors from previous products, thereby reducing costs and improving efficiency in managing the intricate configurations and stringent safety

²¹At the time this work was reviewed.

Table 9. MDE-DevOps-AI/ML selected paper with available tools. Aer. (Aerospace), Aut. (Automotive), D.L. (Digital Life), I. (Application Domain Independent).

Selected Paper	Tool Name	Tool Link	MDE→DevOps	MDE→AI/ML	DevOps→MDE	DevOps→AI/ML	AI/ML→MDE	AI/ML→DevOps	MDE + AI + DevOps	Requirements	Modelling	Coding	Testing	Monitoring	Domain(s)	Case Studies
[SP7]	JTL	https://github.com/MDEGroup/jtl-eclipse	■	-	-	-	-	-	-	-	■	-	-	■	I.	2
[SP10]	ACCORDANT	https://github.com/knilo-castellanos/accordant-usecases	■	■	-	-	-	-	■	■	■	-	-	-	Aer., I.	1
[SP15]	Joda-Time	https://github.com/JodaOrg/joda-time	-	-	-	-	-	■	-	-	-	-	-	-	I.	1
[SP21]	N.A.	https://github.com/sh-t-20/artifacts	-	-	-	-	■	-	-	-	■	-	-	-	Aut.	2
[SP24]	N.A.	https://github.com/TimKam/agent-based-process-orchestration-iot	■	-	-	-	-	■	■	-	-	-	-	-	I.	1
[SP30]	iObserve	https://github.com/research-iobserve	■	-	-	-	-	-	-	-	■	-	-	■	I.	1
[SP36]	AURORA	https://github.com/MDEGroup/AURORA	-	-	-	-	■	-	-	-	■	-	-	-	I.	1
[SP44]	FLiRTS	https://cazzola.di.unimi.it/flirts2.html	■	-	-	-	-	-	-	-	-	-	-	■	I.	0
[SP45]	CloudML	https://github.com/SINTEF-9012/cloudml	■	-	-	-	-	-	-	-	■	-	-	■	D.L.	1
[SP46]	CloudML	https://github.com/SINTEF-9012/cloudml	■	-	-	-	-	-	-	-	■	-	-	■	I.	4
[SP52]	OSTIA	https://github.com/maelstromdat/OSTIA	■	■	-	-	-	-	■	-	-	-	-	■	I.	3
[SP59]	N.A.	https://github.com/SeelabFhdo/seaa2020	■	■	-	-	-	-	-	■	■	-	-	-	I.	2
[SP63]	Lowcomote	https://github.com/lowcomote/	■	-	■	-	-	-	-	■	■	■	■	■	I.	1
[SP71]	RIG	https://ls5download.cs.tu-dortmund.de/rig/	-	-	-	-	-	-	-	-	■	-	-	-	I.	1
[SP77]	Microservice Identification	https://github.com/miguelfbrito/microservice-identification	-	-	-	■	-	■	-	-	■	-	-	-	I.	1
[SP87]	FFSM Diff	https://damascenodiego.github.io/learningFFSM/	-	-	-	-	■	-	-	-	■	-	-	-	Aut., Aer., I.	5
[SP92]	ML-Quadrat	https://github.com/arminmoin/ML-Quadrat	-	■	-	-	-	-	-	-	■	■	-	-	I.	0
[SP98]	DivENACT	https://gitlab.com/enact/divenact	-	-	-	-	-	-	■	-	-	-	-	-	I.	0
[SP100]	DICER	https://github.com/dice-project/DICER	■	-	-	-	-	-	-	-	■	-	-	-	I.	0
[SP104]	MontiAnna	https://www.se-rwth.de/materials/deeplearning/	-	■	-	-	-	-	-	-	■	■	-	-	I.	0
[SP109]	MontiThings	https://github.com/MontiCore/montithings	-	-	■	-	-	-	-	-	■	■	-	-	I.	1
[SP112]	openCAESAR	https://github.com/opencaesar/kepler16b-example	■	-	■	-	-	-	-	■	■	-	-	-	Aer., I.	1
[SP113]	MICO	https://github.com/UST-MICO/mico	-	-	-	-	-	-	-	-	■	■	-	■	I.	1
[SP115]	Pinset	https://github.com/alfonsodelavega/pinset-examples/tree/master	-	■	-	-	■	-	-	-	-	-	-	-	I.	0
[SP116]	CodeBot	https://parallelagile.com/codebot.html	■	-	-	-	-	-	-	-	■	■	■	-	I., D.L.	4
[SP126]	Skyfire	https://github.com/mdsol/skyfire	■	-	-	-	-	-	-	■	■	-	-	■	I.	1
[SP128]	SMADA-Fog	https://github.com/penenadpi/smada-fog	■	-	-	-	-	-	-	-	■	■	-	■	I.	2
[SP131]	ContinuousEvolution	https://github.com/jokingarcia/ContinuousEvolution	■	-	■	-	-	-	-	-	■	■	-	-	I.	1
[SP134]	DockerConflict	https://github.com/tbten/dockerconflict/tree/master	■	-	-	-	-	-	-	-	■	-	-	■	I.	1
[SP150]	Proyecto Grado Uniandes	https://gitlab.com/sergiorom92/proyecto-grado-uniandes	■	-	-	-	-	-	-	-	-	-	-	■	I.	0
[SP153]	ShipYard	https://github.com/lowcomote/shipyard-operational-semantics	■	-	-	-	-	-	-	■	-	-	-	-	I.	1
[SP154]	Lowcomote	https://github.com/lowcomote/	■	-	-	-	-	-	-	■	-	-	-	-	I.	1
[SP162]	Scalable Views	https://github.com/atlanmod/scalable-views-heterogeneous-models	■	-	■	-	-	-	-	-	■	■	-	■	I.	1
[SP170]	OSTIA	https://github.com/maelstromdat/OSTIA	■	■	-	-	-	-	■	-	■	-	-	■	I.	3

requirements in automotive systems. CodeBot and FLiRTS address the need for rapid and flexible solutions in dynamic digital environments for Digital Life applications. CodeBot focuses on automatically generating microservices from domain models, enabling the swift translation of specifications into executable code and supporting agile development cycles. FLiRTS employs fuzzy logic to automate regression test selection based on UML models, which helps reduce both the cost and effort needed to maintain high-quality standards in digital contexts. The remaining tools are application-independent, meaning they are not tied to any specific domain. This flexibility allows them to be effectively applied across various contexts, including cloud computing, IoT, edge and fog computing, and big data analytics, providing versatile modeling and automation solutions.

Finally, it is worth noting that FFSM Diff, CodeBot, CloudML, and OSTIA have been validated through multiple case studies, demonstrating their robustness and effectiveness in real-world environments. These tools have

undergone large-scale empirical evaluations, underscoring the potential of integrated MDE, AI/ML, and DevOps solutions to address complex challenges successfully across industrial and research settings.

4.4.4 RQ2 Discussion. Our analysis of software engineering activities revealed a strong focus on modelling and testing. However, areas like requirements engineering pose significant challenges that could be addressed by approaches identified in this study. These findings emphasize the value of a comprehensive classification of papers, which provides researchers and practitioners with essential resources to enhance their work in specific software engineering activities. The keyword analysis emphasized testing and modelling within the selected papers, reflecting the current research focus in these areas. In contrast, the relatively lower occurrence of coding-related terms suggests that this area may be underrepresented in the current literature. As a consequence, it could offer different interesting opportunities for further research. Our ontology, developed to capture semantic relationships between keywords from the selected papers, includes frequency information indicating the weighted relationships among nodes. Significant correlations among MDE, DevOps, and AI/ML were observed, providing a comprehensive understanding of the current research landscape and highlighting clear areas for further exploration.

To complement this, the performed topic analysis offered a foundational understanding of the content within the selected papers. Notably, it highlighted the pivotal roles of the MDE, AI/ML, DevOps, CPSs, and Cloud domains, suggesting a classification of papers based on their prominence in these areas. While the data corpus structure influences the identified topics, the topic analysis served as an exploratory tool to guide further interpretation and deeper insights by the authors.

Finally, tool availability is a crucial aspect as far as software and systems engineering is concerned. Indeed, our assessment highlighted a mixed landscape regarding tool availability. While many papers make research prototypes or tools accessible, a significant number of them currently lack clear information on tool availability. Such a situation can interfere with replication efforts and the practical application of the presented research findings. Consequently, improving the clarity and accessibility of tool information in these scientific publications could significantly enhance research reproducibility and utility. For example, this is particularly important when combining MDE and DevOps with a data-oriented area such as AI/ML. Future work should provide more accessible tools and more straightforward information about their availability. This is fundamental to support a more robust and usable body of research in software and systems engineering.

4.5 Which research fields and applications domains have been considered (RQ3)

4.5.1 Research fields In addition to the keywords, ontology, and topic analysis, we also analyzed the venues of the selected papers. The objective was to extract additional themes and topics of interest. The venues were analyzed by extracting the conference, journal, and workshop names from the publication metadata. The venues' names were then manually classified into categories based on the research area or topic that each venue covers. The results of the venue analysis are presented in Figure 12 (a). This figure provides the number of papers published in each category and the percentage of papers in each category.

The results of the venue analysis show that most papers in the study were published in journals, conferences, and workshops that focus on “Software Engineering”, “Languages”, and “Model-Driven Engineering”. This is in line with the findings from the other analysis. The venue analysis also identified other research areas and topics that are of interest to the research community. These notably include “Object Computing & Technology” and “Formal Methods”.

An important observation from the venue analysis is the lack of AI/ML-specific venues focused on Software Engineering. While AI/ML is widely applied in SE, the most relevant research appears in SE-oriented venues,

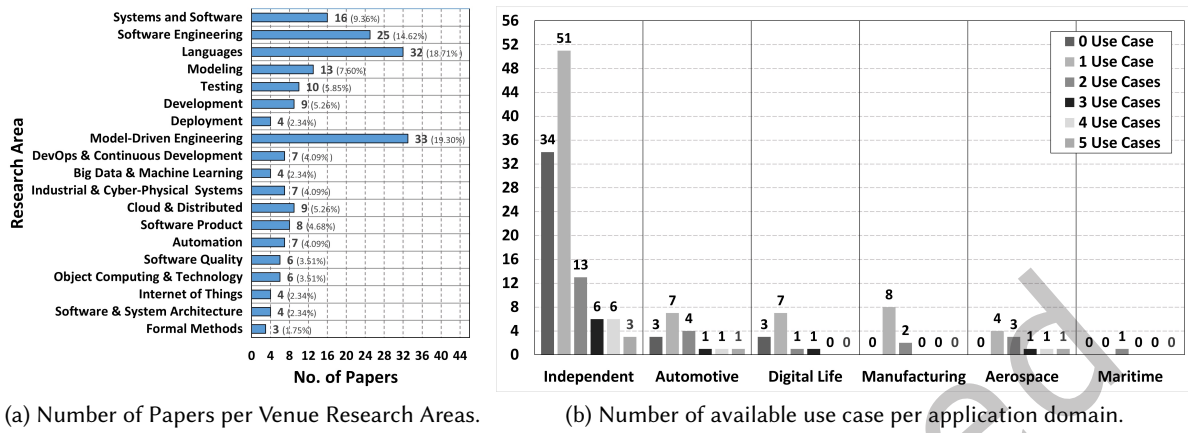


Fig. 12. Venue research areas and use case availability.

suggesting that AI/ML conferences prioritize theoretical advancements over SE applications. This limits interdisciplinary collaboration despite AI/ML's growing role in software development. Bridging this gap may lead to better knowledge transfer and help transfer AI/ML innovations into mainstream software engineering.

4.5.2 Application domains The main application domains and related market figures have already been discussed in previous works [60]. These main application domains are automotive, aerospace, railway, maritime, construction, digital life, and manufacturing.

Figure 12 (b) shows a classification of the contributions from the selected papers according to the application domains. When the contribution of the selected paper is not tied to a particular domain, it is classified as *domain independent*. It is worth noting that each selected paper can contribute or not (i.e., with zero or more contributions) to an arbitrary number of domains.

As a result, the most supported domains are automotive, digital life, manufacturing, and aerospace. None of the selected papers proposes domain-specific solutions for the railway and construction domains. This absence denotes an opportunity for further investigations to explore tailored methodologies and tools for the particular issues that arise in software and system development in these industries.

It is also vital to mention that the majority of the selected papers propose MDE, DevOps, and AI/ML contributions that are not explicitly limited to a specific application domain, thus suggesting that they can be reused or at least inspirational for researchers and practitioners in the field of continuous software and system development.

Since many application domains integrate AI/ML, fostering cross-disciplinary venues could help bridge this gap. Future work should explore ways to increase SE visibility in AI/ML communities and encourage collaboration between researchers in both fields. This will lead to more focused and practical AI/ML solutions for complex software development problems.

4.5.3 Case Study/Example availability In this section, we further analyze the presence of available examples in the selected papers. We specifically focus on use case papers and their relation to application domains. The obtained results are presented in Figure 12 (b). This figure shows the distribution of papers by the number of included use cases (ranging from 0 to 5) according to the CPS application domains

Our analysis reveals that papers with one use case are the most common (77 occurrences). This is followed by papers with no use cases (40 occurrences), corresponding to works that introduce new concepts without

real-world validation. Additionally, 24 papers present 2 use cases, while 9, 8, and 5 papers present 3, 4, and 5 real use cases, respectively. This analysis provides insights into the types of examples used in the literature and the application domains they represent. These insights can inform future research directions in this area.

4.5.4 RQ3 Discussion. The analysis of venues confirmed a strong focus on already well-established areas such as software engineering, languages, and model-driven engineering. This observed situation is consistent with the continued relevance of these fields in developing software modelling tools. Interestingly, the analysis also revealed a growing interest in related areas such as data computing and formal methods. Analyzing the application domains within the selected papers revealed a more varied landscape. Automotive, digital life, manufacturing, and aerospace emerged as the most prevalent domains of interest. However, we observed a lack of research papers focusing on the railway or construction domains, for example. This distribution suggests potential future research directions that would explore the application of modelling solutions within these currently underrepresented sectors. It is important to note that many of the selected papers presented approaches that applied MDE, DevOps, or AI/ML to non-specific domains. This finding highlights the potential for such approaches to be more generic and applied across multiple domains. Finally, the use case analysis revealed that most papers considered a single use case, with a notable number not considering any. While the papers without use cases generally focused on introducing novel concepts or innovative ideas, this finding highlights the importance of relying on more real-world validation through use cases in future research.

4.6 Future research directions (RQ4)

This section discusses some open research directions that emerged from this study.

- **Classifying AI/ML techniques in synergies with MDE and DevOps.** Integrating AI/ML with MDE and DevOps can offer improved data analysis [SP1], enhanced model development processes [SP121], and extend streamlined software engineering processes and platforms already integrating MDE and DevOps aspects [SP153]. Therefore, investigating the AI techniques utilized in the selected papers and creating subcategories based on these techniques could be an interesting avenue for future research.
- **Bridging the gap of AI/ML synergies with both MDE and DevOps.** One of the primary outcomes of this study is the significant under-representation of synergies between AI/ML and both MDE and DevOps. In the work by Cabot et al. (2018), the authors envision potential collaborations between AI/ML and MDE, highlighting that AI/ML techniques require extensive knowledge bases to produce reliable results. For example, in MDE, the lack of large, curated datasets containing well-classified examples of “good” and “bad” artifacts poses a considerable challenge, limiting the training potential of these algorithms. In the context of AI and ML models, having a clean and curated data set allows for making accurate predictions and better decisions, which in turn fuels the models themselves. The quest to maintain the quality and availability of data is an essential part, highlighted in many of the selected papers in this study (e.g., [SP29, SP119, SP143]), which will require coping with a diverse set of information sources and legacy systems. Furthermore, as Casale et al. (2015) point out, successful synergies between AI/ML, MDE, and DevOps necessitate data-aware methodologies to effectively manage development and operational stages. To address this issue, we propose guidelines and recommendations to bridge the gap between AI/ML technologies and MDE and DevOps.
 - **Need for open data repositories.** In general, the availability and management of domain-specific data still need to be improved. Ideally, the required improvement can be achieved through well-established, community-driven repositories that provide high-quality data for domain-specific tasks. The objective would be to provide the required data to train the next generation of domain-specific AI tools. To this end, the implementation or adoption of standardized data-sharing schemes, e.g., *Schema.org* extensions like Croissant [5] or DCAT [19], can be considered. This would notably enable data and web crawlers to discover available datasets when progressively released by the community automatically. Moreover,

the MDE and DevOps communities must invest in creating and maintaining curated training datasets tailored specifically to these fields' unique needs. Identifying the specific tasks inside the domains that would benefit from prospective AI tools (e.g., model transformation in MDE) should lead to the design of these particular datasets. Once again, regular updates to these datasets would be crucial for reflecting current trends and ensuring the robustness of corresponding AI/ML applications.

- **Historical perspective and challenges.** Creating, developing, and maintaining relevant data repositories over the years is challenging. For example, in the MDE community, several initiatives have already been launched since the year 2000 to build such a repository (cf. ReMoDD [29]) or at least lists of related platforms and tools²². However, despite significant effort from the research community, they generally failed to achieve great success due to the lack of clear long-term incentives (notably from an economic perspective). We also observed that companies are generally reluctant to publicly share many of their domain-specific modeling artifacts (models, transformations, generators, etc.) because these artifacts provide them with a competitive advantage. As a result, the open data in MDE is often limited to generic models, simple transformations, or toy examples.
- **New opportunities with Machine Learning.** We believe that the recent advent of ML techniques, particularly Large Language Models (LLMs), could significantly help reduce these past and current limitations. Thus, we are likely to witness an increase in contributions related to AI/ML integration within MDE and DevOps in the near future. LLMs could assist practitioners in generating and curating the necessary datasets. This is notably true for MDE, in which LLMs could be used to produce large collections of models with specific structures and characteristics, e.g., for benchmark purposes. As a concrete example of a solution that provides the kind of bridge connecting MDE and AI, CNNGen [33] offers a DSL to describe CNN architectures. It allows you to automatically generate Python scripts to train and evaluate the modeled architectures from these descriptions (i.e., models). Such CNN architectures could then be used to populate target data repositories.
- **Supporting scalability.** DevOps, MDE, and AI/ML focus on scalability. The practices and principles of these areas play a crucial role in improving the scalability and management of high-demand applications or services. DevOps principles such as infrastructure as code and containerization facilitate the scalability of software systems; similarly, ML frameworks and cloud platforms enable the training and deployment of scalable models. Even in the MDE area, much attention was paid to scalability and flexibility. However, scalability is still a critical factor in this context (e.g., [SP19, SP143]), as managing vast amounts of data and adapting to fluctuations and increasing complexity is often necessary.
- **Continuous software engineering and continuous learning.** While DevOps and MDE focus on continuous integration, deployment, and monitoring, AI/ML adds complexity through continuous learning. As models evolve, their behavior changes dynamically, raising challenges like model drift, stability, and explainability. Maintaining AI-driven systems requires real-time monitoring and retraining, but avoiding unintended consequences remains challenging. Software engineering can help with model validation, automated testing, and traceability, as well as adapting techniques like automated versioning, runtime verification, and explainability frameworks. Future research should explore how MDE and DevOps can enhance continuous learning, ensuring stability and maintainability. AI-aware DevOps (e.g., MLOps for SE) could help detect and mitigate model drift, improving software reliability in AI-powered systems.
- **Software Variability.** A promising research direction is the integration of software variability modeling with ML feature engineering. Variability modeling, widely used in Software Product Line Engineering (SPLE) and MBSE, systematically manages configurable software artifacts, while feature engineering is key in optimizing ML models. Recent studies suggest that bridging these approaches could enhance deep learning applications

²²Curated list of UML tools: <https://modeling-languages.com/uml-tools/>

[59]. For instance, Ghamizi et al. [31] propose a feature model of Convolutional Neural Networks (CNNs) in which each valid configuration corresponds to a fully deployable and trainable CNN. They automatically map feature selections to TensorFlow/Keras code, employ a diversity-driven search over the configuration space, and show on MNIST and CIFAR-10 that this method can discover architectures on par with handcrafted designs. Casting DNN design as a variability problem thus exemplifies how SPLE and MDE techniques can systematically explore and optimize deep learning design spaces. We expect such cross-disciplinary approaches to raise practitioners' awareness of synergies between AI, MDE, and DevOps, and to inspire further advances in the coming years. Future research could explore how AI/ML techniques can refine variability modeling and how software variability concepts can contribute to more effective feature selection and adaptation in ML pipelines.

- **New MDE methodologies for Continuous Software Engineering.** Continuous software development has become the heart of software engineering. However, the term “continuous” in software engineering encompasses numerous activities, including continuous delivery and integration, and continuous architecture and improvements [28]. In several selected papers, the authors expressed interest in improving or completely covering the different dimensions of the continuous. Many selected papers aim, as future work, to focus on the improvement or full coverage of different dimensions of the continuous. For example, continuous model-based architecture and continuous evaluation of related changes in the architecture [SP41], model-driven methodology for continuous delivery of infrastructure in software in the cloud [SP8], MDE for continuous integration [SP132], and continuous adaptation [SP5]. We believe that adapting MDE methodologies to meet recent continuous engineering needs would take a significant next step toward the maturity of model-driven engineering practices.
- **DevOps for IoT and Cloud.** IoT ecosystems typically include various components, including sensors, actuators, gateways, cloud platforms, and edge devices. These components must work together seamlessly to collect data, process it, and take action in real time. Although DevOps was originally designed for traditional software development, its focus on collaboration, communication, and task automation can be adapted and expanded to meet the specific challenges of IoT development [SP109, SP62, SP53]. Integrating AI and ML into IoT systems requires sophisticated DevOps practices to manage and update AI/ML models effectively. Such areas of research can impact trustworthiness in IoT systems (integrity, reliability, etc.), and the evolution from “monitoring” to “supervision” can improve feedback and provide developers not only with detailed information but with recommendations and steps to resolve system anomalies and failures [SP62].
- **Digital Twin for DevOps.** A digital twin serves as a comprehensive software representation of a physical system, which includes structures, properties, conditions, behaviours, history, and possible futures of that system through models and data to be continuously synchronized[25]. It provides real-time insights and enables software development, testing, and deployment. Several selected papers argue about the use of digital twins for supporting continuous monitoring, evolution, adaptation, and verification (e.g., [SP166, SP157, SP144, SP65, SP68, SP165, SP121]).
The digital twin can significantly contribute to overcoming DevOps challenges; we envision several benefits of implementing a digital twin in DevOps: (a) comprehensive testing and validation of software behaviour before deployment, identifying and addressing potential issues early in the development lifecycle (e.g., [SP146, SP144, SP145, SP157, SP165]); (b) facilitated real-time monitoring and feedback loops (e.g., [SP146, SP144, SP161, SP62, SP97, SP157, SP65, SP165]); (c) increased system reliability and continuity, and minimized disruptions in DevOps pipelines caused by issues (hardware or software) that can be detected early by the digital twin (e.g., [SP106, SP144, SP157, SP55, SP165, SP89, SP151, SP31]); and (d) improved scalability, reliability and reproducibility of DevOps processes for testing, deployment and production (e.g., [SP145, SP78, SP4, SP144, SP157, SP165, SP151]).
- **MDE, AI/ML, and SecDevOps.** DevOps is gaining more and more success; however, there remains a risk that such fast and continuous integration and delivery processes introduce additional security vulnerabilities

into systems in production. SecDevOps [36] has arisen in response to this major security problem, promising to bridge the gap between continuous release cycles and security needs by addressing security at every development lifecycle stage. Several selected papers proposed approaches to support security in DevOps (e.g., [SP169, SP145, SP124, SP121, SP53, SP121, SP134, SP144]).

We believe in both model-based methods and intelligent techniques. By leveraging security processes as a core element of DevOps pipelines, we can support the efficient and continuous engineering of industrial systems.

5 Quality Evaluation

In this research, we adhered to established guidelines for systematic studies. However, our work may still be subject to specific validity threats. Below, we outline these potential threats and the strategies employed to mitigate their impact.

5.1 Construct Validity

Construct validity threats concern the accuracy of deriving correct conclusions from the relationship between treatment and outcome [65]). Several measures have been taken to deal with threats to construct validity. For example: (i) avoid mono-operation bias by selecting the reviewers based on different backgrounds (MDE, DevOps, AI/ML research areas) and research experience, (ii) adopt accurate methods and protocols for all the steps of the review process (spanning from databases selection through research questions formulations to and data/results analysis). However, additional improvements are still possible, such as:

- A thorough discussion of extracted results to ensure that all the members of the team share the same understanding of the tasks,
- Including an exhaustive forward and backward *snowballing* search in the protocol to ensure that no important contribution is missing,
- Ensuring that papers match reviewers' areas of expertise to provide a high-quality review
- Taking into account potential conflict of interest, anonymity, and confidentiality guaranteed in the papers review process to avoid evaluation apprehension.

Finally, the search string shown in section 3.3.1 enabled making this SMS feasible at this stage, as it focused on the search of papers fully aligned with AIDOaRt research interest (in the sense of being likely to consider the three research axes). However, it also introduced a bias in the search for works answering any of the three former research questions RQ1, RQ2, and RQ3, involving only two of the three axes.

5.2 Internal Validity

Internal validity threats relate to poor settings impacting the study design [66]. Exclusion and randomisation have been included to address the threats to internal validity related to the confounding variables (a typical potential source of bias in one empirical study). For the exclusion criteria, we have selected papers based on the three research areas (MDE, DevOps, AI/ML) by excluding ones that, for example, (i) have not been published (e.g., Bachelor, Master, and PhD thesis) or (ii) published from low ranked journals (higher than Q1/Q2 Scimago) and conferences (GRIN-GII-SCIE).

We have selected papers targeting at least two of the three research areas (i.e., we excluded tool papers mentioning only 1 of the 3 research areas, i.e., MDE, AI/ML, and DevOps). We intend to further improve such classification of the papers by refining the selection based on their contribution to the three research areas. As shown in Figure 5 and Figure 6, we have selected papers that both propose well-assessed techniques for MDE, AI/ML, and DevOps and adopt their basic principles.

Moreover, to avoid the well-known problem of selection bias, we have used a random sample of potential search strings (for all three research areas). We have also addressed issues related to the ambiguity and incompleteness

(poorly worded) of the RQs (as well as the search strings and inclusion/exclusions criteria) in different ways, such as: (1) we have iteratively reviewed the RQs (search strings and inclusion/exclusions criteria) during several meetings with consortium's experts; (2) our search strings are based on well-assessed search strings (typically used by well-defined existing empirical studies focused on one of the three research areas, MDE, DevOps, AI/ML); (3) we have performed the papers review process by using several measures (e.g., (i) meetings for making the reviewers familiar with the review form or (ii) different interactions for assessing the quality of the review process, such as we have piloted the study results by analyzing them in different iterations). We intend to continue working to address additional threads related to the ambiguity of the data extraction forms to help reviewers avoid potential inconsistencies.

5.3 External Validity

External validity threats refer to generalizing causal findings to the intended population and settings [66].

We have performed an automatic search on leading research and rich databases to provide adequate literature coverage. The databases were selected based on well-assessed literature guidelines and suggestions; therefore, we assume that the results apply to large-scale studies. Our study mainly investigated whether existing scientific literature supports at least two of the three research areas (MDE, DevOps, and AI/ML). Further extensions (aimed at boosting the generalization of our results) could relate to, for example, considering other review criteria (such as particular domains or aspects/methods of the MDE, DevOps, and AI/ML paradigms). To address the research questions of this study, inspired by the Software Engineering BoK, we have used the MDE BoK, the AI BoK, and the DevOps BoK to provide a detailed classification of the literature. This classification also enables to capture and illustrate a careful mapping of the technical areas across the three fields. However, choosing a different classification framework may result in different views over the same papers, which could be appropriate for addressing other research questions.

5.4 Conclusion Validity

Conclusion validity threats affect the relationships between the extracted data and obtained findings [66]. We have used several measures in all the steps of our study to ensure that our conclusions are correct (e.g., a solid protocol for RQs definition and statistical analysis methods for evaluating the results).

5.5 Exploitation

This work provides an overall context for researchers to know if their contributions are in a mature and competitive area or on the opening of a new one. Also, the industry can be interested in the immediate exploitation of the mature areas while monitoring the recent ones.

In this respect, this work also provides clues on which application domains these fusions have been oriented. This way, the researcher is aware of whether the target application domain of its approach is crowded with competitors. If not, it can help to reconsider whether targeting that application domain is interesting.

The research also provides information on other practical aspects, like whether use cases and tools are available in the area, allowing researchers to use them as a baseline for improving methods and reproducing comparable experiments. The availability of commercial or open-source tools is also relevant information for developers and use-case stakeholders.

Finally, the work also provides information to researchers interested in the fusion of two or more of the addressed research areas (MDE, AI/ML, DevOps) to address the specific papers dealing with it among the selected paper list through additional materials³.

6 Conclusion

This paper presents an overview of state-of-the-art research approaches and solutions that rely on different synergies of MDE, AI/ML, and DevOps to support various software and systems engineering activities. Because of the large number of available resources from the literature, we performed the study and presented the obtained results as a Systematic Mapping Study (SMS). To the best of our knowledge, this is the first SMS to address all the synergies between the three areas considered.

As a consequence, the main objectives of our SMS are multiple: 1) Identify software and systems engineering methodologies or approaches that explicitly combine principles and techniques of MDE, AI/ML, and DevOps, and explain how they are currently combined mostly two by two. 2) Study which software and systems life cycle steps are concerned by these methodologies and approaches, and to what extent. 3) Further analyze the particular research fields and application domains targeted by these methodologies and approaches. 4) Extend this analysis by providing a mapping between corresponding tools and targeted engineering activities, highlighting their role within the identified MDE, AI/ML, and DevOps synergies. 5) Propose relevant future research directions for the software and systems engineering community to explore, notably related to integrating these three considered areas.

The results of our SMS show that few approaches explore the synergies between MDE, AI/ML, and DevOps to support software and systems engineering. However, many approaches already combine two of these areas, specifically MDE and DevOps. We also highlighted the increasing frequency of approaches exploring synergies between AI/ML techniques with MDE or DevOps. The selected approaches generally cover various activities, such as requirements engineering, modeling, coding, testing, and monitoring, to different extents. This is realized thanks to corresponding tools, which mostly focus on such two-by-two synergies for particular engineering activities in specific application domains (even though some identified tools aim to be more generic). Still, many research challenges remain open. Potential research directions notably include pushing further the actual integration of AI/ML with MDE and DevOps. They also include supporting more scalable and data-oriented approaches, proposing new global methodologies for continuous software and systems engineering, and adapting DevOps practices more efficiently to different systems (e.g., IoT- and/or Cloud-based, Digital Twins).

From our side, we intend to continue improving regularly the SMS to address additional threats to validity (related, for example, to results generalization) and integrate new relevant scientific papers whenever they are published. This SMS has been designed to be an open/live artefact, shared with the software and systems engineering community, to be possibly refined and updated by different interested researchers. Additionally, we plan to gain further insights by conducting developer questionnaires and incorporating the findings into the current study to assess the applicability of our SMS results for industry practitioners.

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